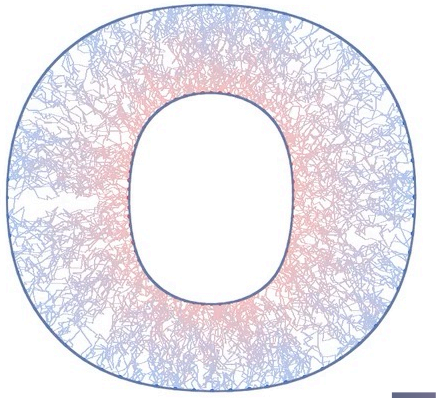
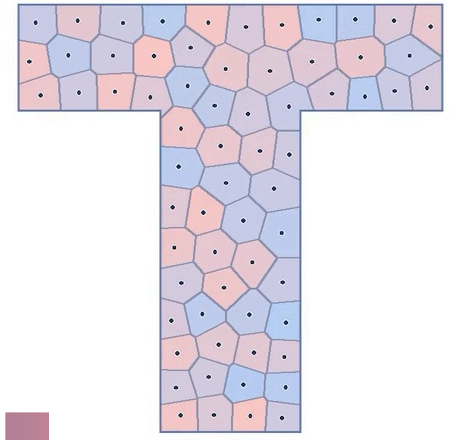
# Invited Lecture: Intro to Optimal Transport

With FOCUS ON APPLICATIONS

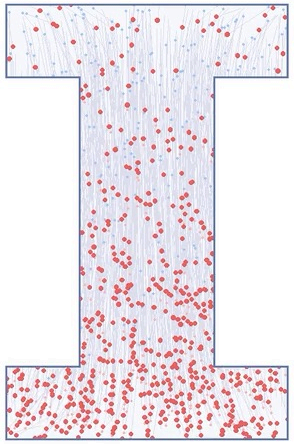


Entropic bridges

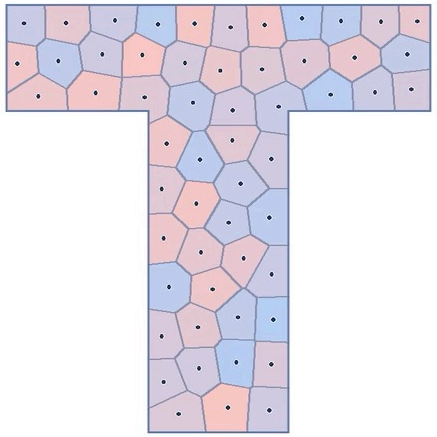


Lloyd quantization

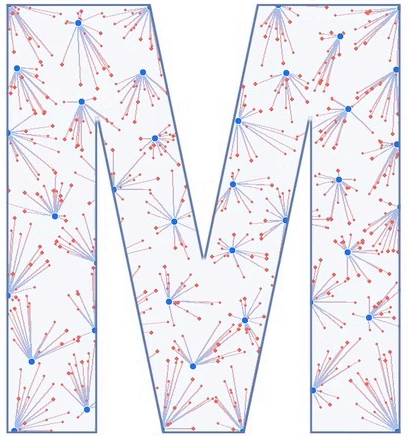
IN



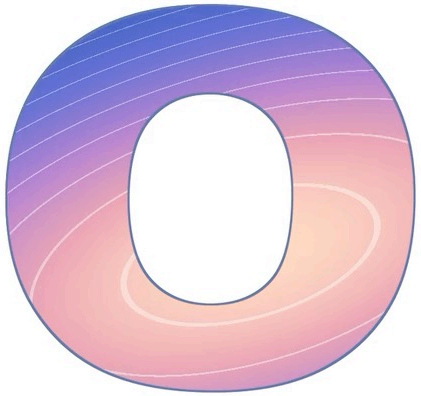
Displacement



Biased quantization



Monge transport

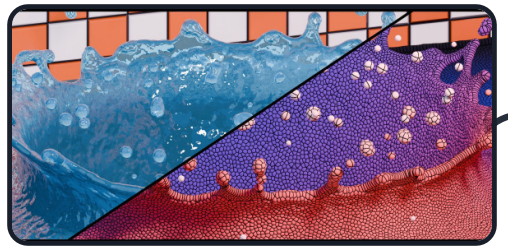


Wasserstein barycenter

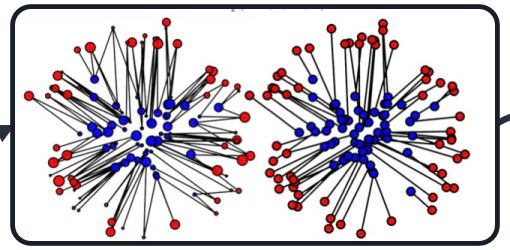
# Optimal Transport: Matching, Generating and Beyond

Click on block to jump straight to the topic!

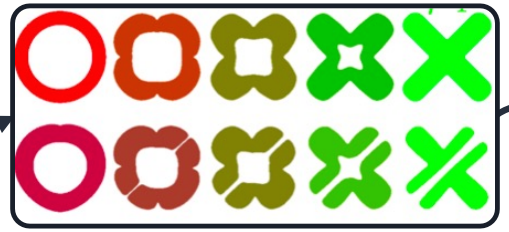
## Why OT?



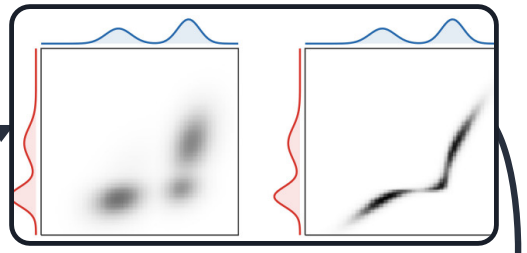
## Maps & Couplings



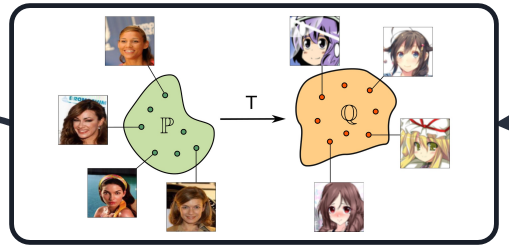
## Wasserstein Geometry



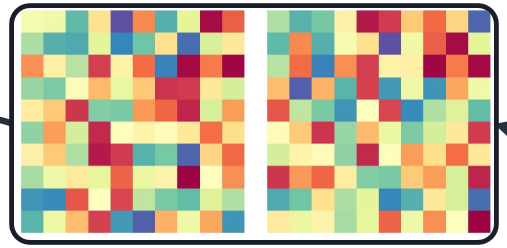
## Computing OT



## Generative Transport

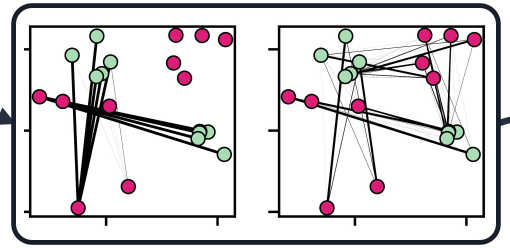



## Model Alignment

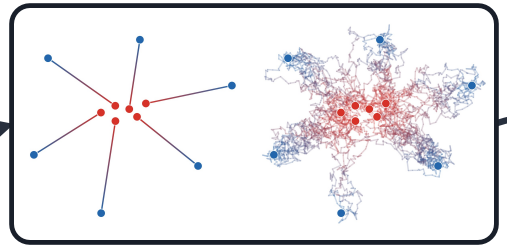


## OT in Fluid Dynamics

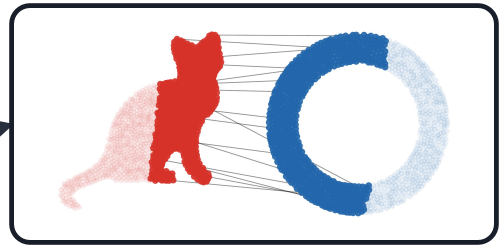
## Mixing & Routing



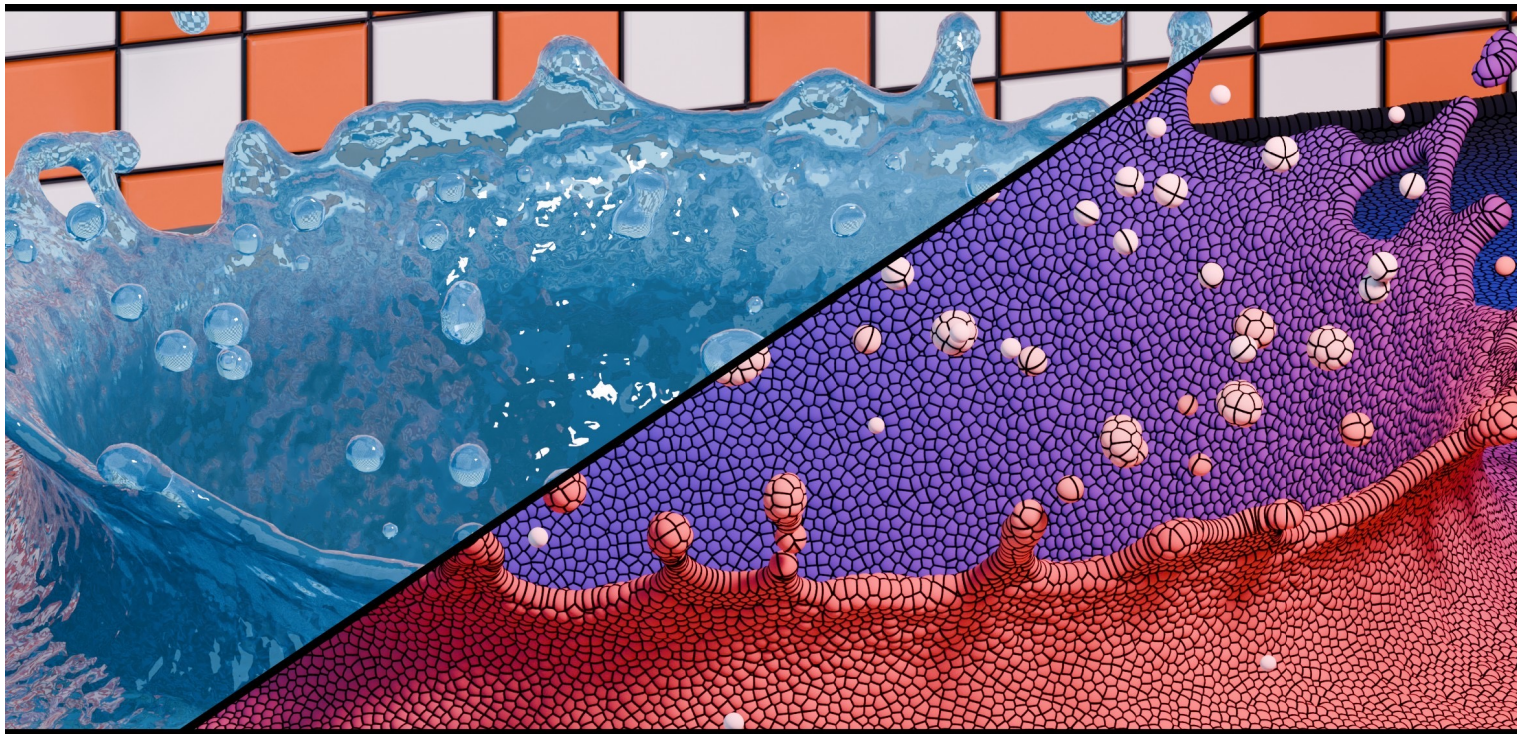
## Schrödinger Bridges



## Beyond Classical OT



# Why OT?



# Model Merging for Image Generation

**Used in Stable Diffusion image-generation pipelines!**

**Pokémon fine-tune** + **Sketch fine-tune** = **Merged model**

$$\theta_{\text{merge}} = (1-\alpha)\theta_A + \alpha\theta_B$$

## Why merge?

Blend learned styles

One checkpoint at inference

Merge without retraining

## Linear Mode Connectivity

1. Sinkhorn coupling (neuron costs C)

$$P^\star = \arg\min_{P\in\Pi} \left[\langle C, P\rangle - \varepsilon H(P)\right]$$

2. Average aligned weights

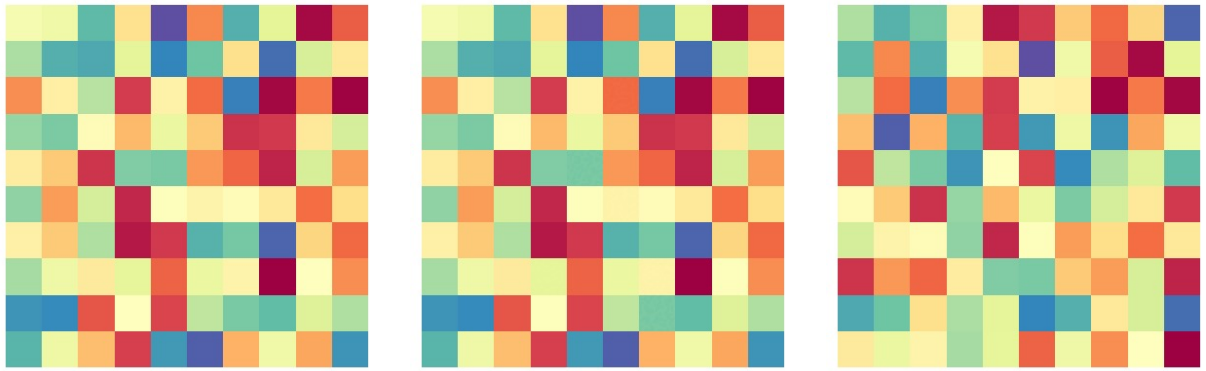
$$\theta_{\text{merge}} = (1-\alpha)\theta_A + \alpha\,\pi_P(\theta_B)$$

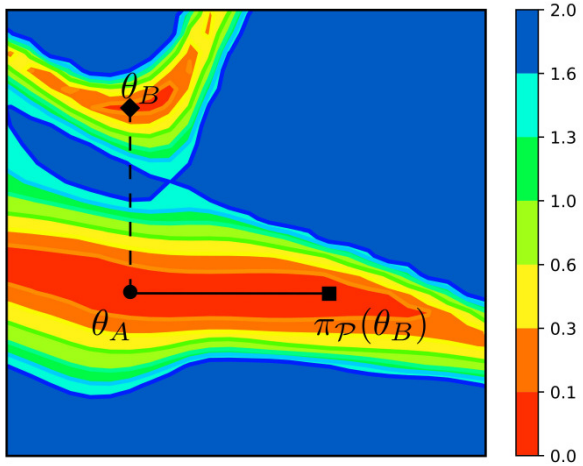
Biggs et al., Diffusion Soup, ECCV 2024, Fig. 5. Peña et al., CVPR 2023.

# OT for Model Merging

## Neuron alignment in a toy MLP

**Network A** | **Reordered A** | **Network B**





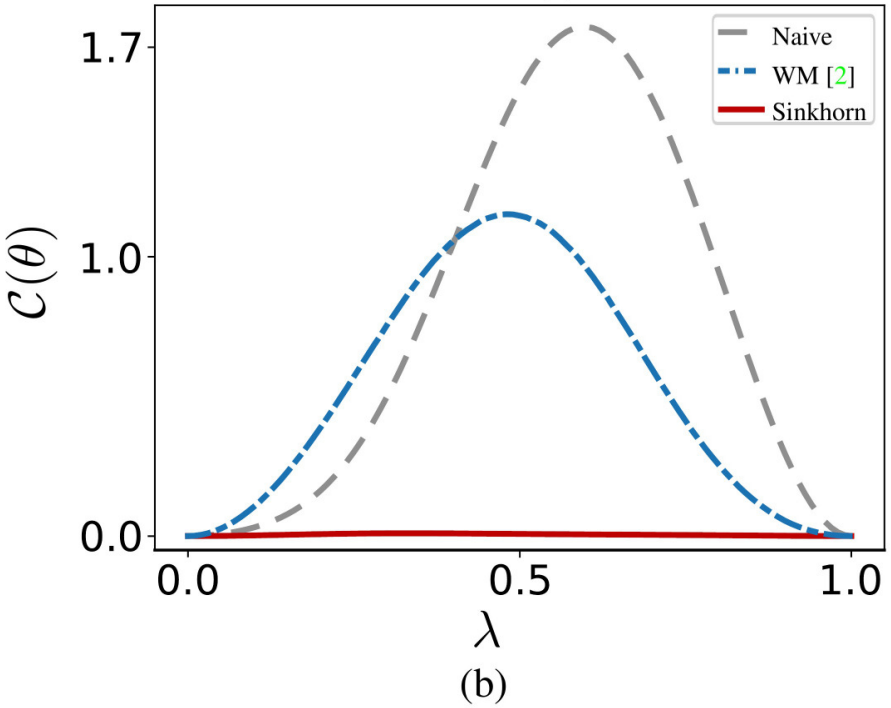




Neuron permutations preserve the network function.

Sinkhorn learns a soft matching before neuron permutation.

## Weight interpolation after alignment




**Alignment can make weight averaging useful.**

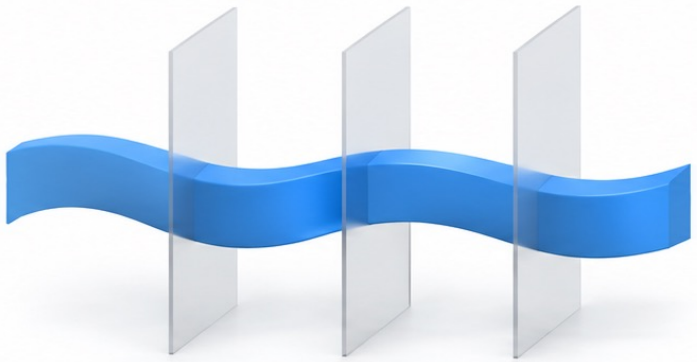
Peña et al., Re-basin via implicit Sinkhorn differentiation, CVPR 2023

# OT in manifold Constrained Hyperconnections (mHC)

**Multiple residual streams give information more paths through depth**

## Residual connection

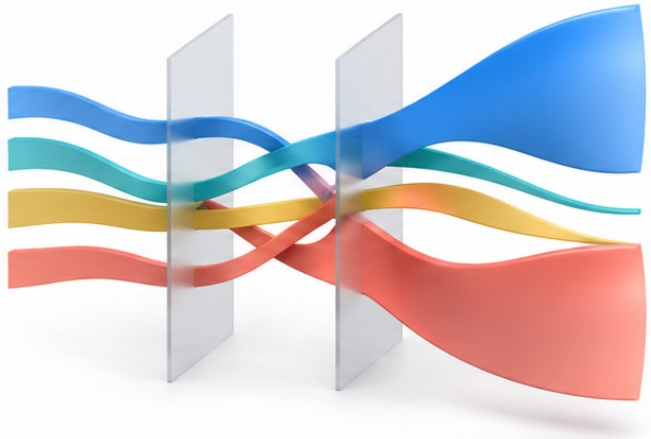
One evolving state



Carry one state through layers.
Add each layer's update.

## Hyper-connections

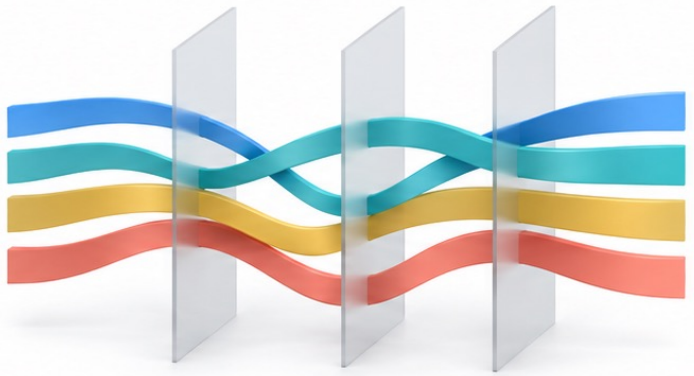
Several interacting states



Mix several streams.
Signals can grow across layers.

## Constrained HC (mHC)

Flexible, controlled exchange



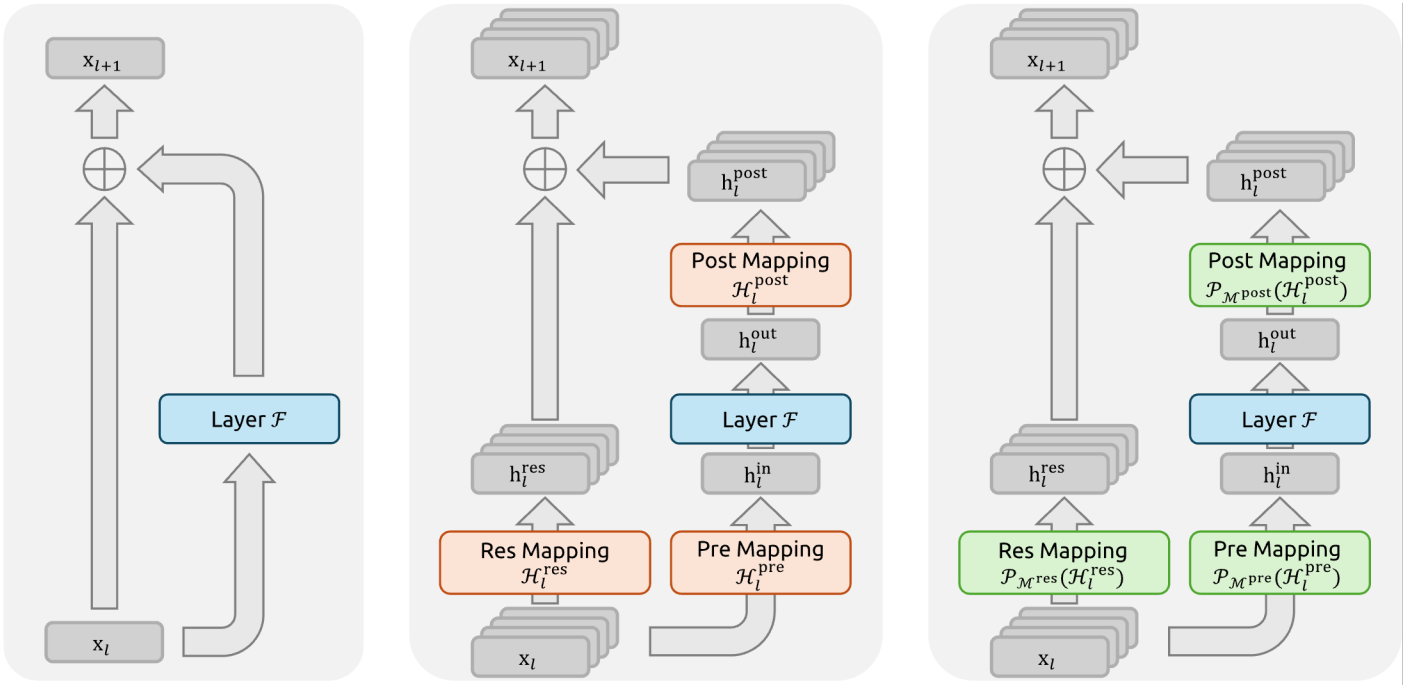
Keep the multiple streams.
Use Sinkhorn to control mixing
Make Doubly-Stochastic

**The goal: richer information flow with stable residual mixing**

Zhu et al., Hyper-Connections (2024) · Xie et al., mHC (2025)

# OT Geometry in DeepSeek mHC

## Residual connection paradigms




(a) Residual Connection (b) Hyper-Connections (HC) (c) Manifold-Constrained HC (*m*HC)

### Unstable mixing

Unconstrained stream mixing can amplify signals across depth.

### Sinkhorn normalization

Normalize the residual mixer toward unit row and column sums.

### Controlled mixing

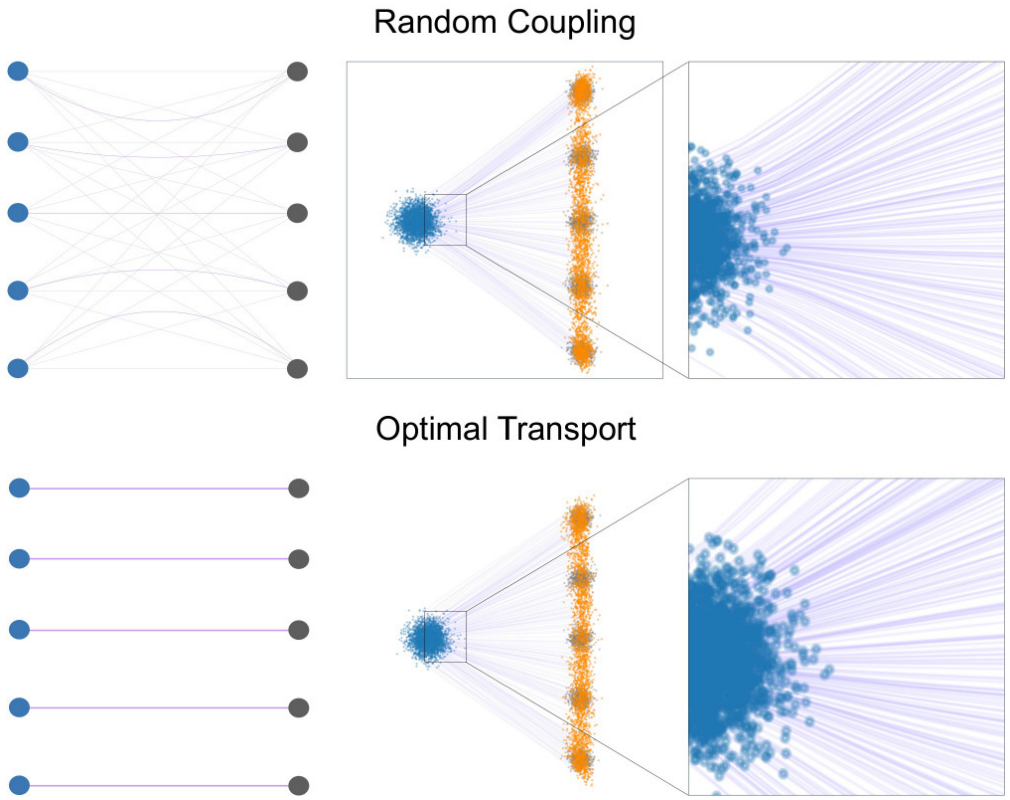
Doubly stochastic mixers stay doubly stochastic when multiplied.

## Repeated residual mixing

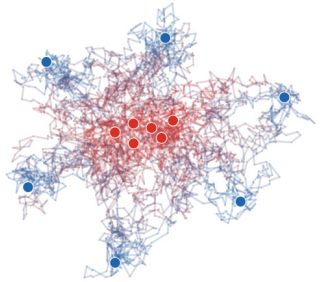
$$X_L = \underbrace{H_{L-1}^{\text{res}} \cdots H_l^{\text{res}}}_{\text{many matrix multiplications}} X_l + \cdots$$

The dots collect the sublayer updates added along the way.
Xie et al., mHC: Manifold-Constrained Hyper-Connections (2026).

# OT for Generative Modelling




OT-CFM (Tong et al.) and COT-FM (Chiang et al., CVPR 2026)



## Schrödinger bridges

Noisy paths connecting two distributions

## Random pairings

Random noise–data pairings can create curved learned flows.

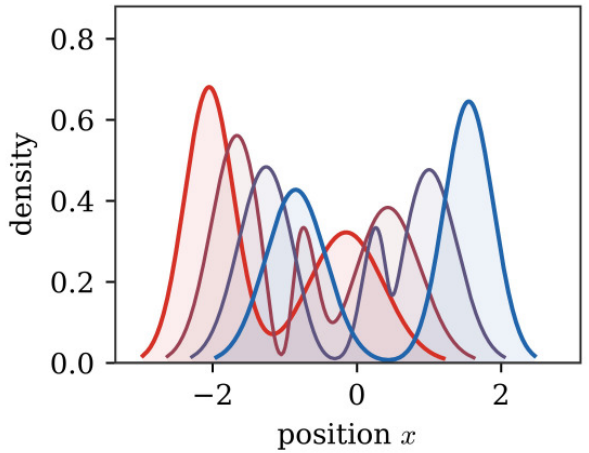
## OT coupling

Match noise and data samples using minibatch OT.

## Straighter flows

Can reduce sampling error with fewer steps.

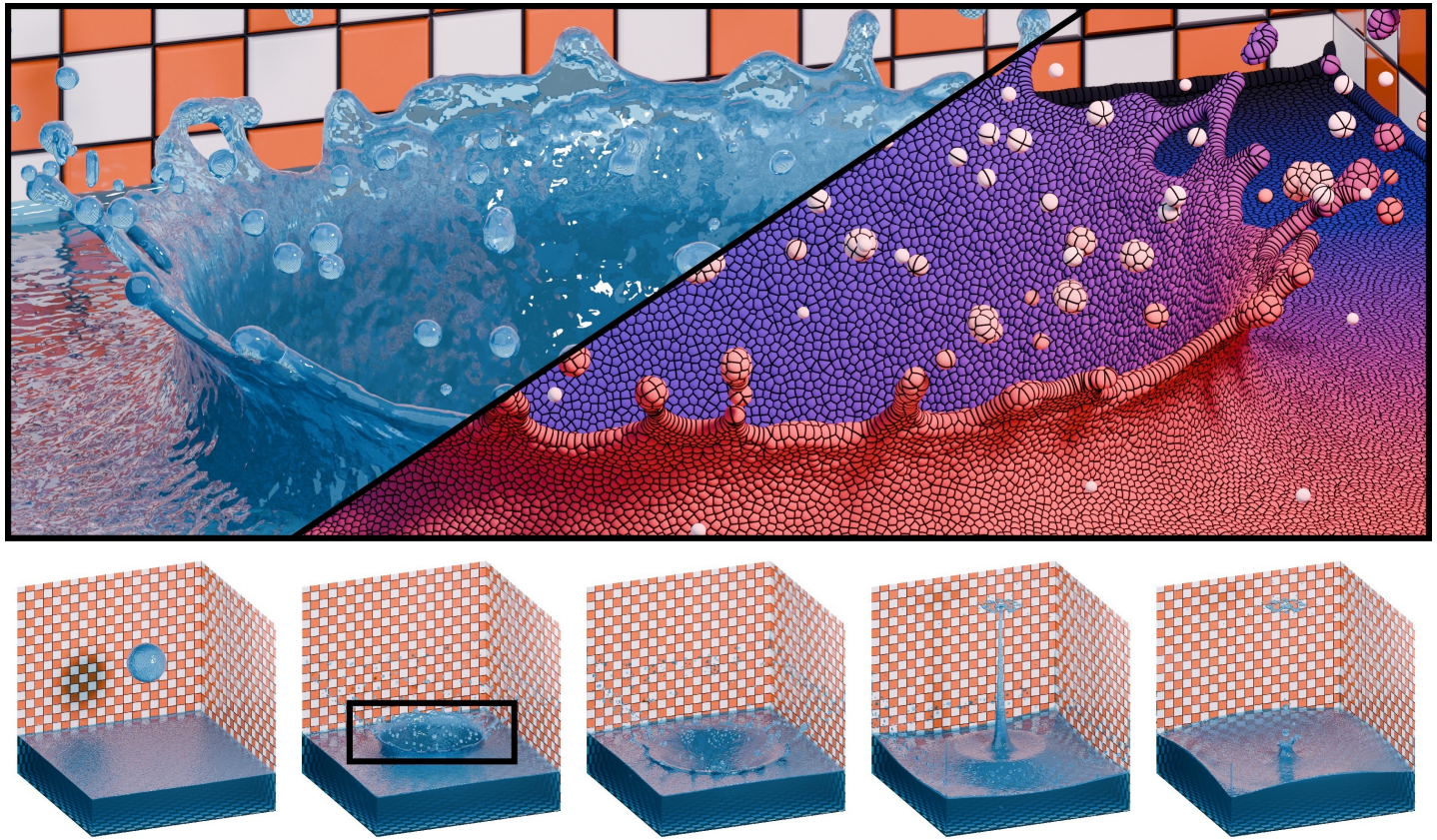
## Continuous density transport




Red: source  Purple: intermediate  Blue: target

# OT for Fluid Simulation

## How can a splash change shape without losing volume?



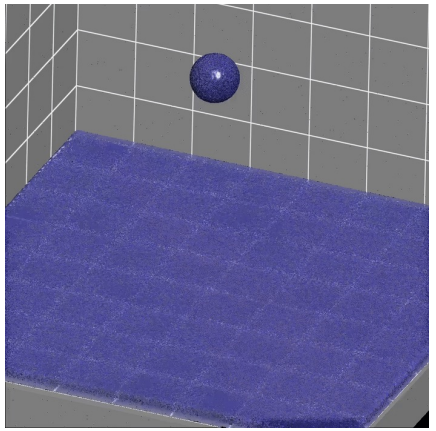
Rendered water and its OT mesh, with 2 million fluid cells

Plateau-Holleville & Lévy, More Power to the Particles (2026)

### Prescribed volume for every particle

OT partitions the water into cells with prescribed volumes.

$$\mathrm{Vol}(L_i) = \frac{m_i}{\rho_0}$$



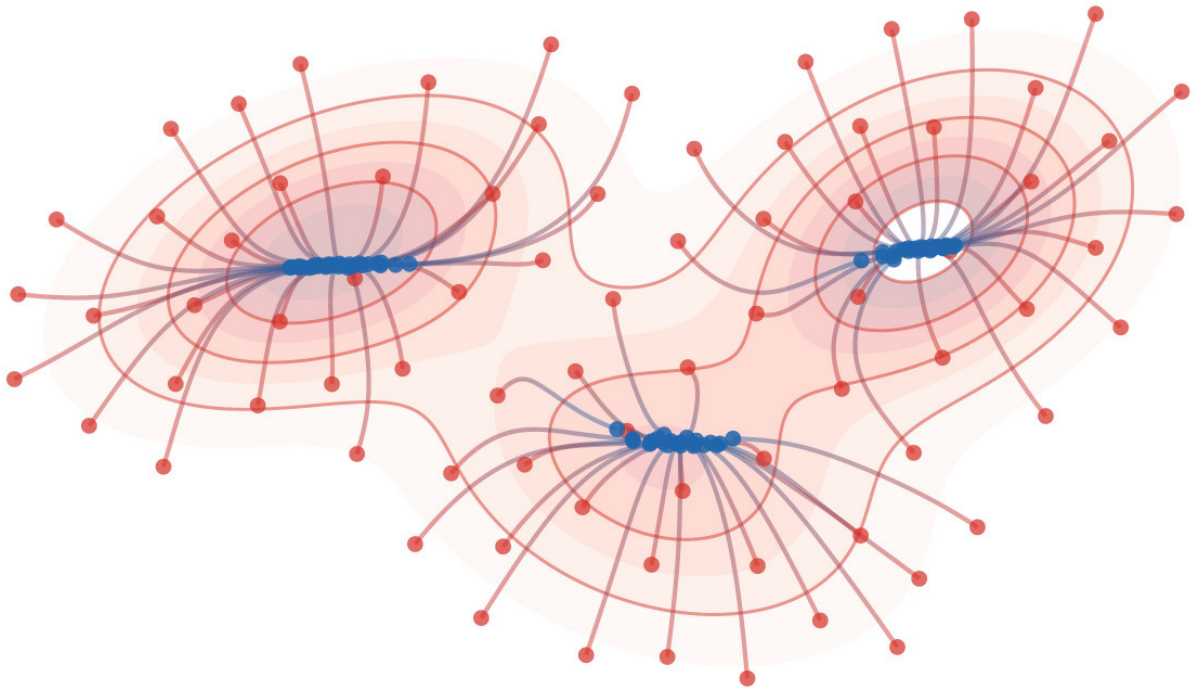
Bruno Lévy, Graphite demo

# Transformer Depth as Transport

## Self-attention moves token vectors in representation space

In “river bank”, context from “river” updates the vector for “bank”.

**One point = one token vector**



Red: early states  Paths across layers  Blue: later states

### Token vectors are particles

Attention updates each vector using weighted information from the context.

### Layer depth acts as time

Small residual updates trace paths through representation space.

### The cloud is a distribution

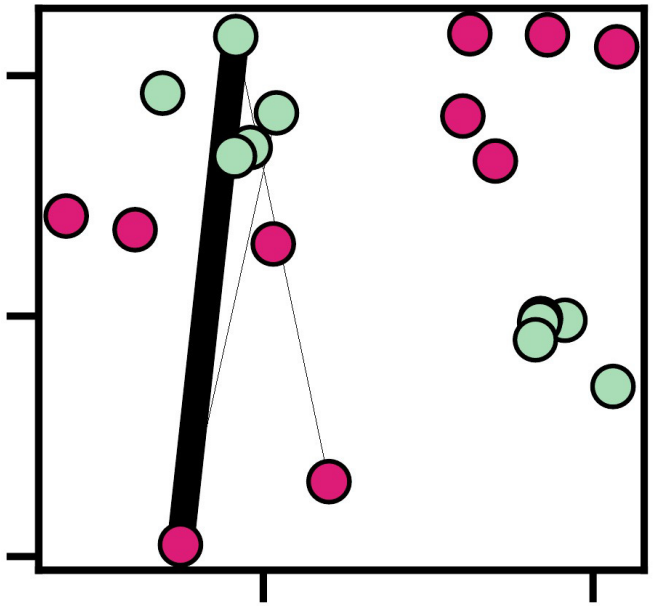
Transport tools study clustering and when token vectors lose distinction.

Transport dynamics. A standard Transformer need not minimize an OT cost.
Castin et al. (2025).

# OT for Balanced Attention

**Unnormalized scores**

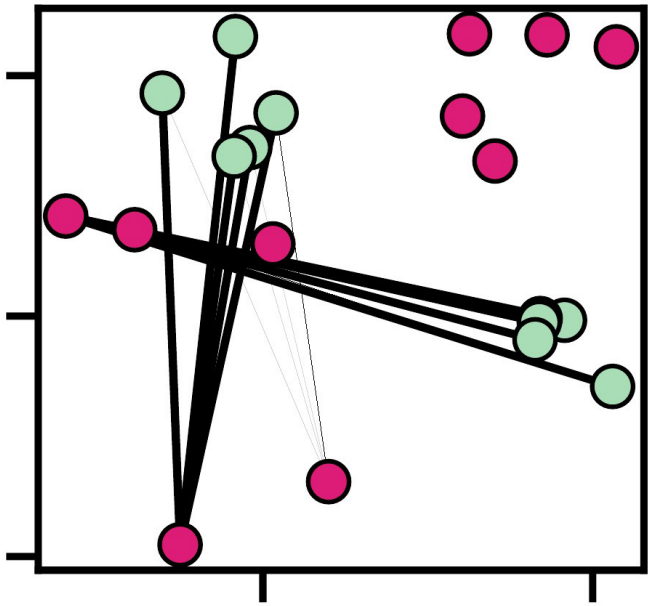
$K^0$



Green = Query
Pink = Key

**Softmax attention**

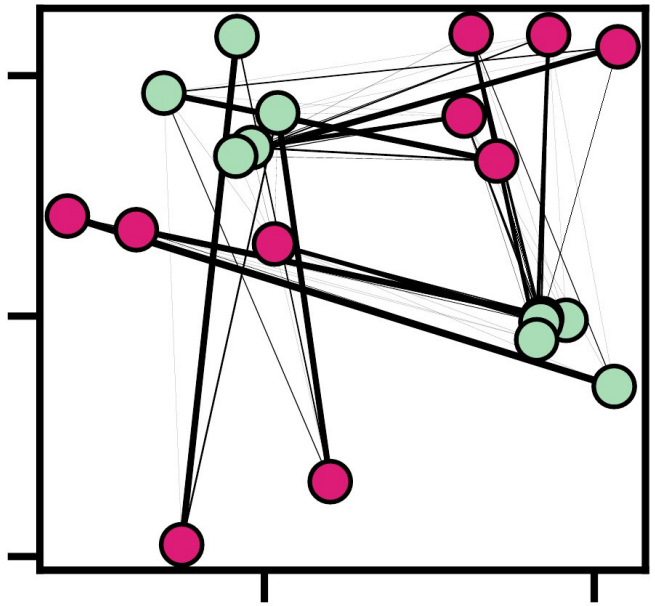
$K^1$



Each query distributes one unit of attention.

**Sinkhorn attention**

$K^\infty$



Both queries and keys have prescribed attention budgets.

Sander et al., Sinkformers: Transformers with Doubly Stochastic Attention, AISTATS 2022

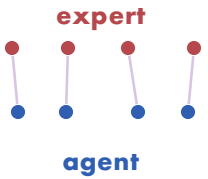
# Optimal Transport in Reinforcement Learning

## OT supplies a geometric signal for imitation, planning and value learning

### Imitation from demonstrations

Door task

Match visits




Match visits to the expert.
OT cost becomes
an imitation reward.

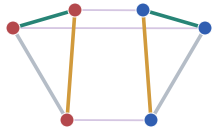
PWIL · Dadashi et al., ICLR 2021

### Gromov–Wasserstein imitation

Expert: Cheetah

Agent: Walker

Relations

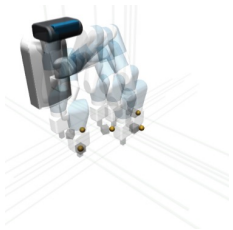


Imitate across state spaces.
Match internal distances.
Turn the match into a
reward.

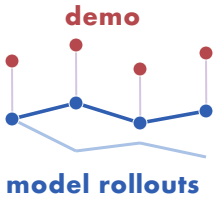
GWIL · Fickinger et al., ICLR 2022

### Zero-shot imitation and planning

Fetch overlay



Score futures

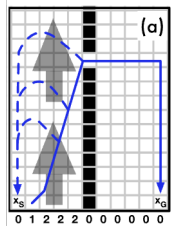



A world model predicts
futures.
OT scores the whole rollout
against a new demonstration.

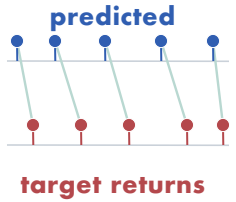
ZILOT · Rupf et al., ICML 2025

### Distributional reinforcement learning

QRTD grid




Match quantiles

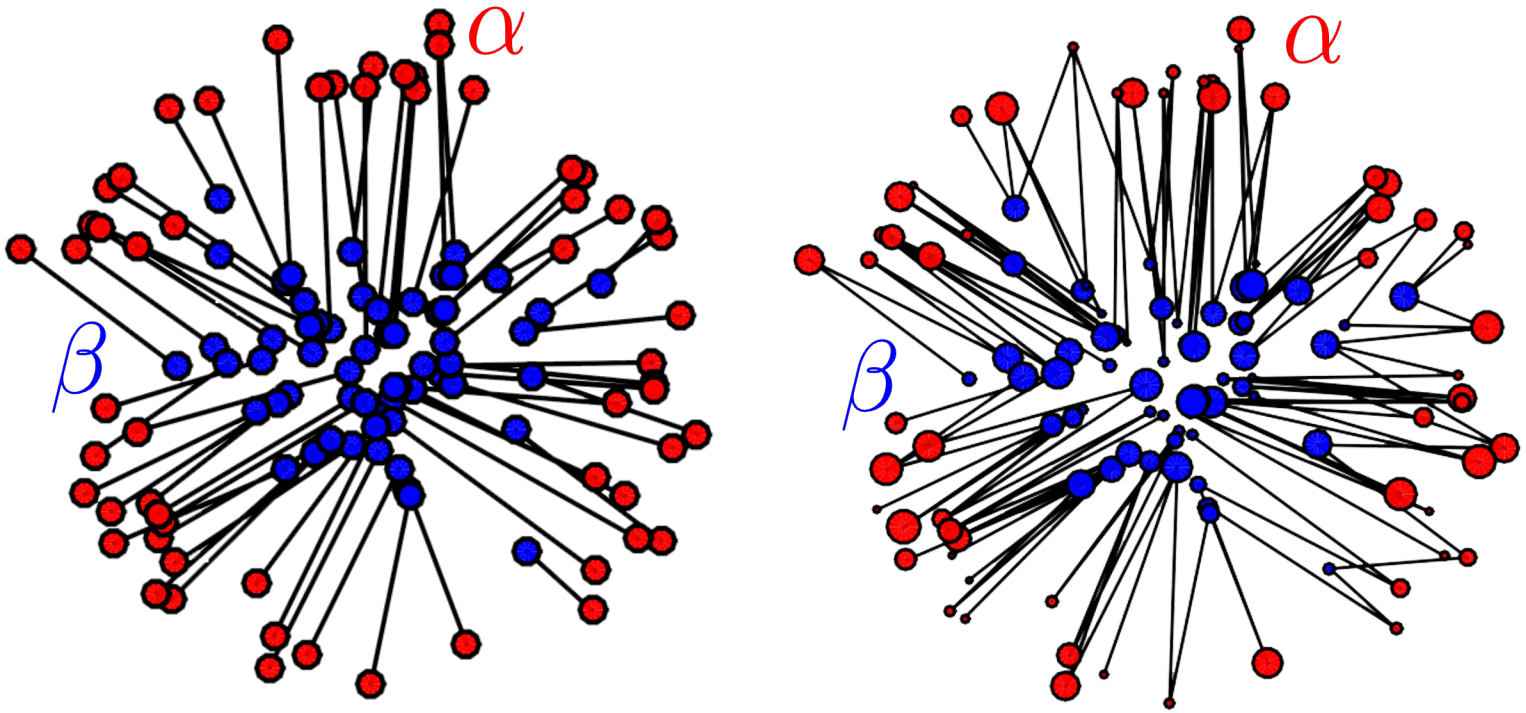



Model random returns.
Learn their quantiles with
Wasserstein geometry.

Quantile RL · Dabney et al., AAAI 2018

**Choosing the ground cost defines which behaviors or outcomes count as similar.**

# Maps & Couplings



# Optimal Transport Between Measures

## Discrete measures

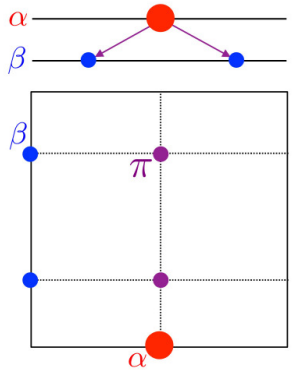
$$\mu = \sum_{i=1}^{n} a_i \, \delta_{x_i}$$

## Measures with a density

$$\mathrm{d}\mu(x) = \rho(x)\,\mathrm{d}x$$

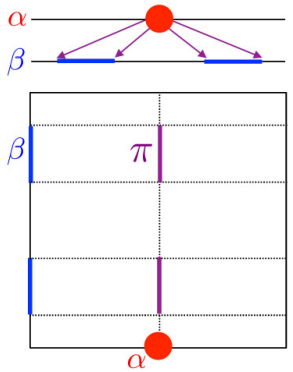
Nonnegative mass. Total mass 1.

### Discrete → discrete



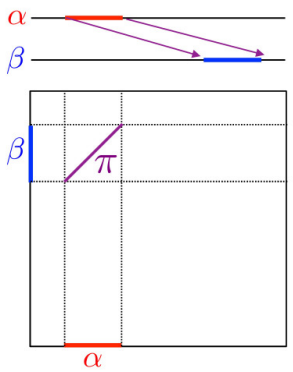

Weights and sample counts may differ

### Discrete → continuous



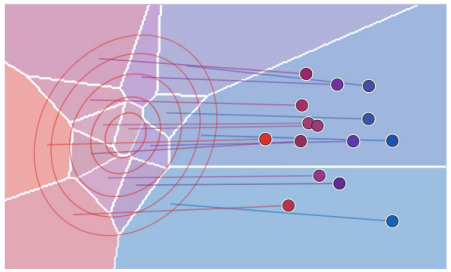

Requires mass splitting

### Continuous → continuous




Uniform densities on intervals

### Continuous → discrete



Regions feed target atoms (Semi-Discrete)

## Monge: transport maps

$$\inf_{T_\sharp \mu = \nu} \int c\big(x, T(x)\big)\, d\mu(x)$$

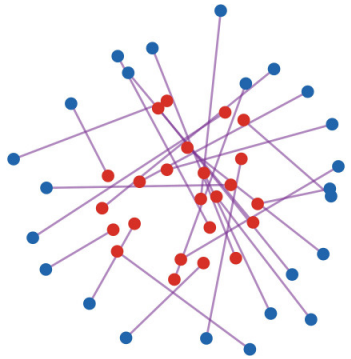
## Kantorovich: couplings

$$\inf_{\pi \in \Pi(\mu,\nu)} \int c(x,y)\, d\pi(x,y)$$

Monge may be infeasible when mass must split.

# Monge OT: One Destination per Point

**A map sends each source point to one target point.**



$$T_{\#}\alpha = \beta$$

**Generally non-convex**

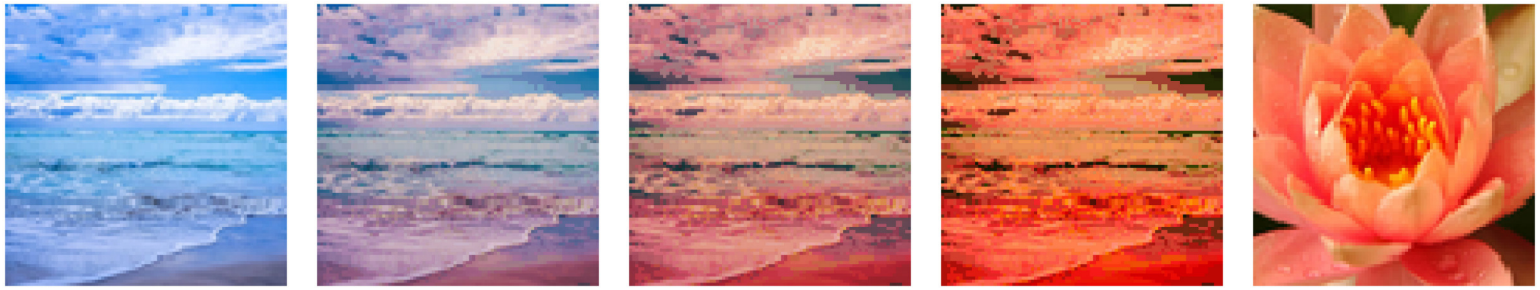
Averages of feasible maps can violate the target law.

**n equal-weight points per measure**

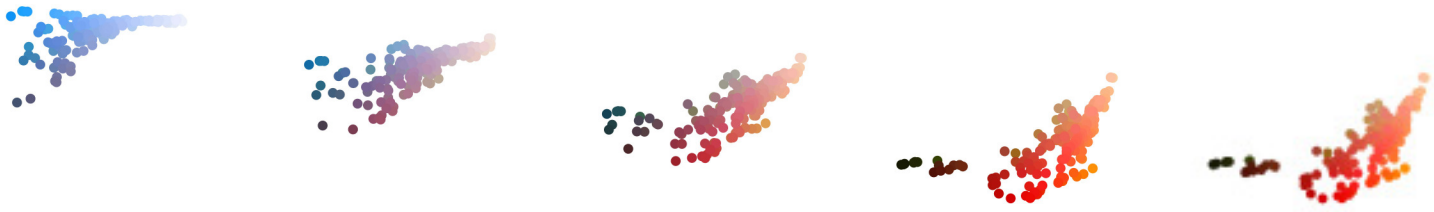
| $n!$ | $\mathcal{O}(n^3)$ |
|---|---|
| possible maps | Hungarian solver |

## Color transfer in RGB space

**Source** **Early** **Midway** **Mapped** **Target**



Empirical RGB color distributions



Colors change while pixel locations stay fixed.

$$\inf_{T:\, T_{\#}\alpha=\beta} \int_{\mathcal{X}} c(x, T(x))\, \mathrm{d}\alpha(x)$$

# Kantorovich OT: Mass Can Split

**Birkhoff polytope**

Equal uniform weights

$nP \in \mathcal{B}_n$

Doubly stochastic after scaling

Vertices: permutation matrices.

$$\min_{\pi \in \Pi(\alpha,\beta)} \int_{\mathcal{X}\times\mathcal{Y}} c(x,y)\,\mathrm{d}\pi(x,y)$$

$$\Pi(\alpha,\beta) = \{\pi \in \mathcal{P}(\mathcal{X}\times\mathcal{Y}) :\ \pi_{\mathcal{X}} = \alpha,\ \pi_{\mathcal{Y}} = \beta\}$$

**DENSE DISCRETE OT**

n points per measure

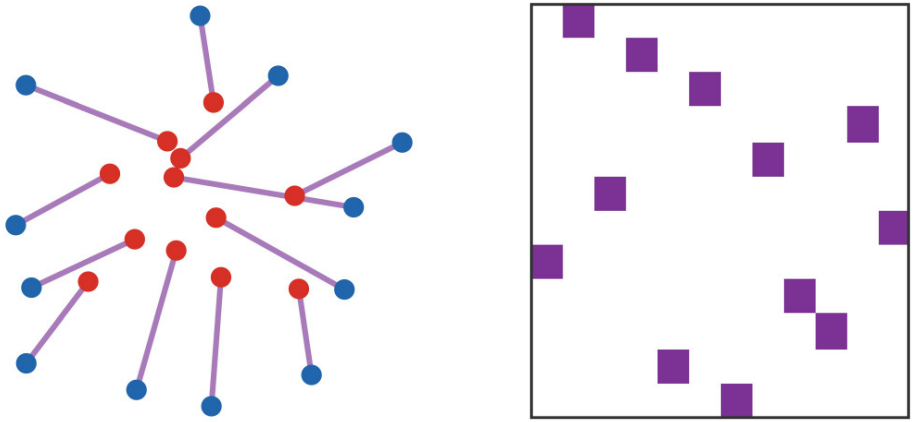
$\mathcal{O}(n^3 \log n)$

Classical min-cost flow

n-by-n transport table

**Convex relaxation: linear cost and linear marginal constraints.**

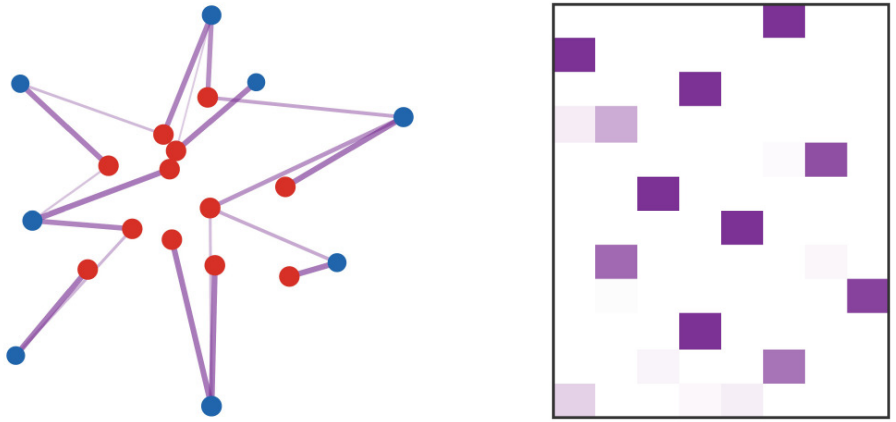
## Uniform weights, equal counts



One destination per source point

$$\pi = \sum_{i,j} P_{ij}\,\delta_{(x_i,y_j)}, \qquad P_{ij} \geq 0$$

## Different target weights

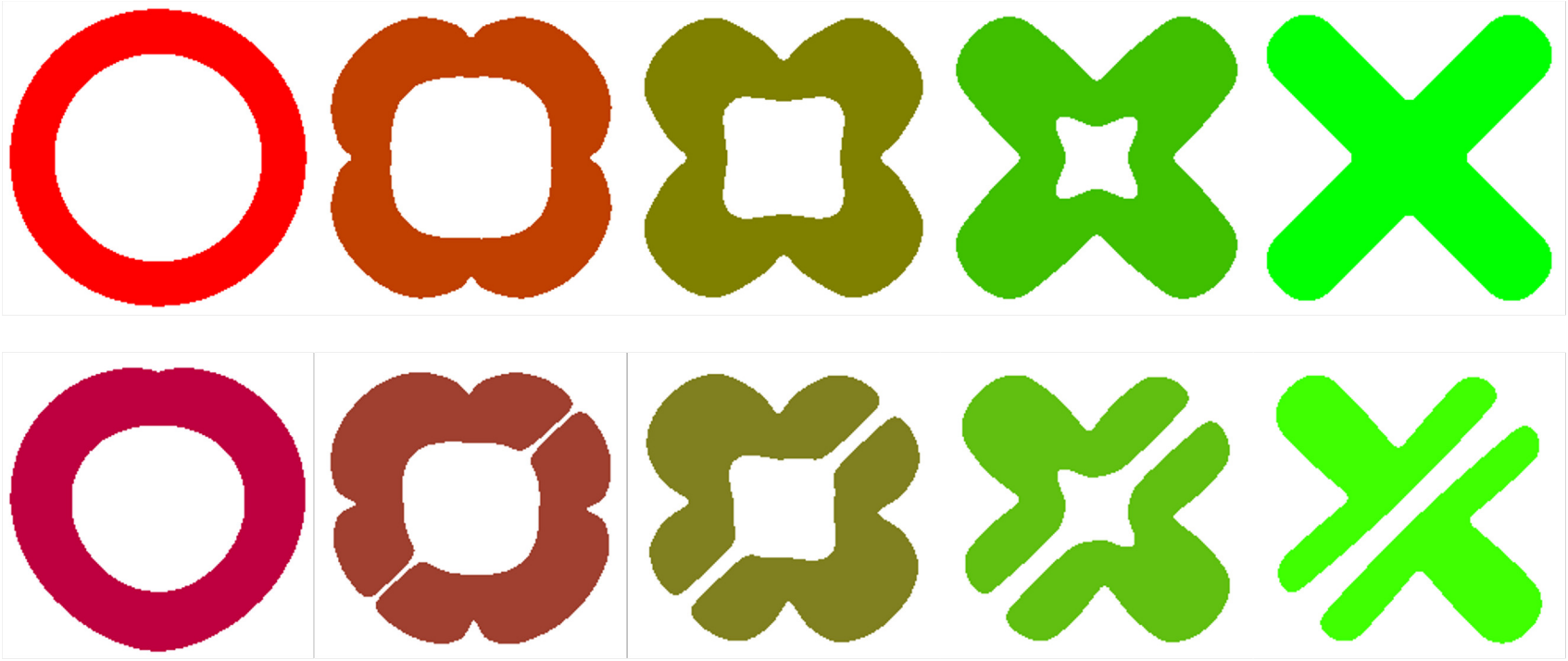


One source can feed several targets

**Mixtures of feasible plans stay feasible**

$$(1-t)\pi_0 + t\pi_1 \in \Pi(\alpha,\beta), \qquad t \in [0,1]$$

# Wasserstein Geometry

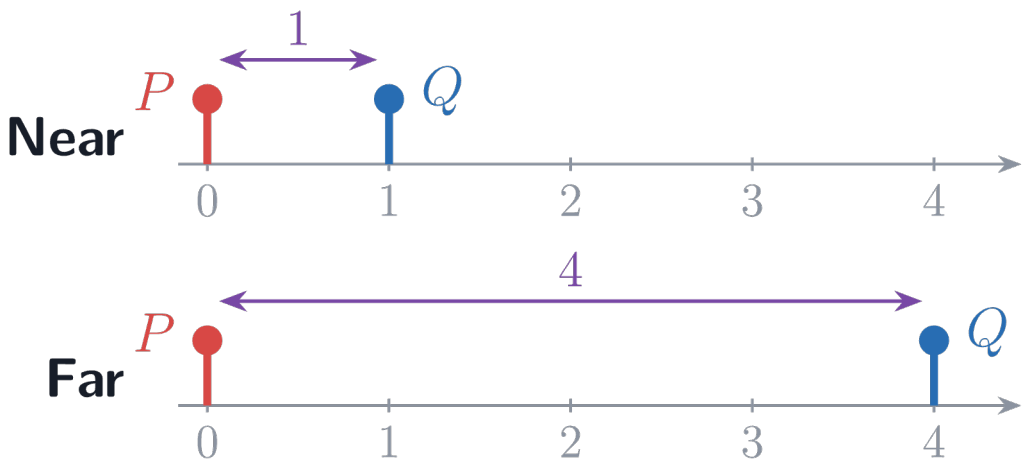


# Wasserstein Distance: Geometry Between Laws

$$W_p(\alpha,\beta) = \left( \inf_{\pi \in \Pi(\alpha,\beta)} \int d(x,y)^p \, \mathrm{d}\pi(x,y) \right)^{1/p}$$

Common metric space; p at least 1; finite p-th moments.

## Near and far point masses




$$W_p(\delta_x, \delta_y) = d(x,y)$$

$$W_p(\alpha,\beta) = W_p(\beta,\alpha), \qquad W_p(\alpha,\gamma) \le W_p(\alpha,\beta) + W_p(\beta,\gamma)$$

## How the measures compare

Same support mismatch; different separation.

| | Near: $Q = \delta_1$ | Far: $Q = \delta_4$ |
|---|---|---|
| $W_p(\delta_0, Q)$ | $1$ | $4$ |
| $\mathrm{JS}(\delta_0, Q)$ | $\log 2$ | $\log 2$ |
| $\mathrm{KL}(\delta_0 \Vert Q)$ | $+\infty$ | $+\infty$ |

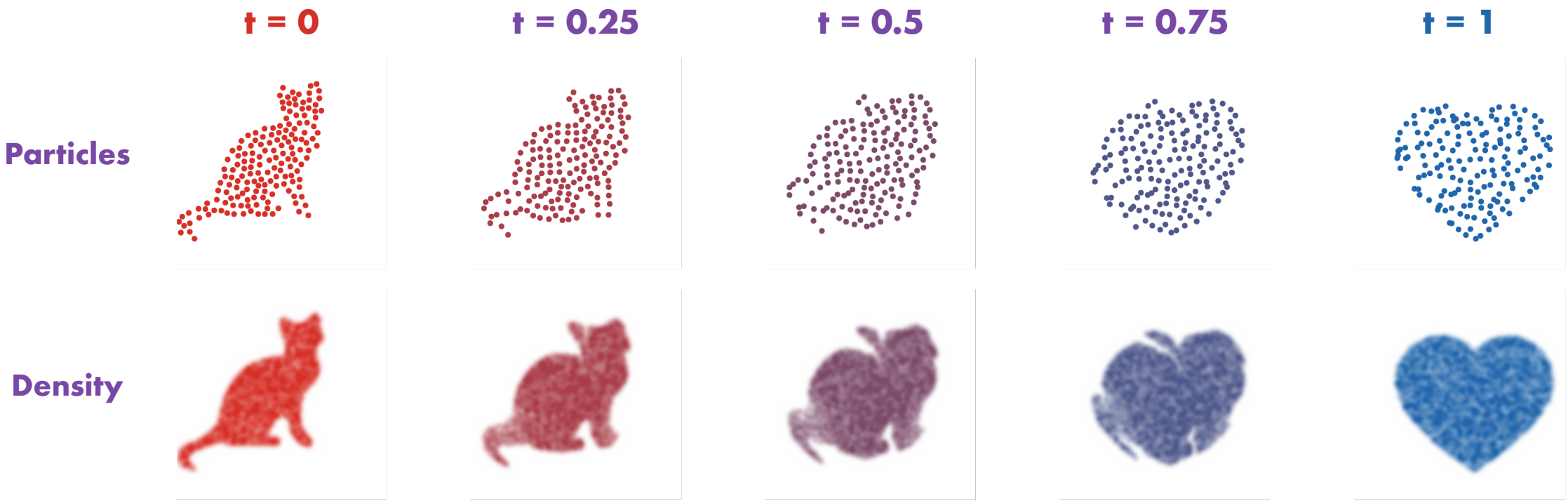
**Wasserstein captures how far mass moves.**

Disjoint supports. Standard JS, natural logs.
Arjovsky et al. (2017), §2.

# Displacement Interpolation

$$(X, Y) \sim \pi^*, \qquad Z_t = (1-t)X + tY, \qquad \alpha_t = \mathrm{Law}(Z_t)$$

A mixture changes weights. A displacement path moves matched mass.

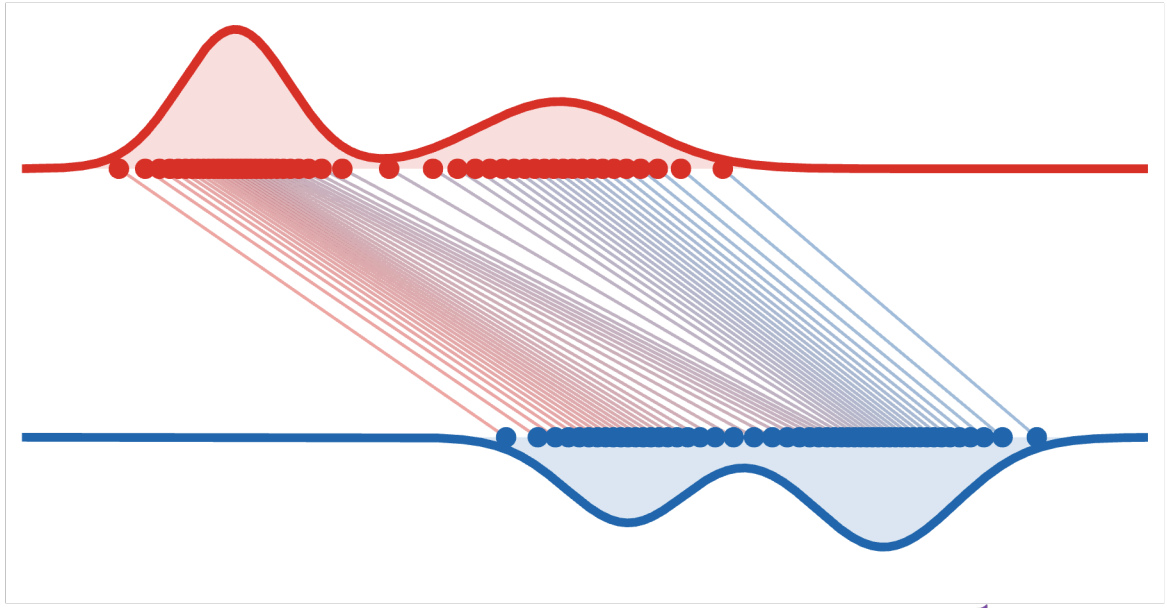



$$W_2(\alpha_s, \alpha_t) = |t-s|\, W_2(\alpha_0, \alpha_1)$$

# Quantile Matching in One Dimension

$$W_p^p(\alpha, \beta) = \int_0^1 |Q_\alpha(r) - Q_\beta(r)|^p \, \mathrm{d}r$$

## Match equal probability levels



## The quantile is the inverse CDF

$$Q_\alpha(r) = \inf\{x : F_\alpha(x) \geq r\}$$

Take the point at probability level r in each distribution and pair the two.

## A geodesic is linear in quantiles

$$Q_t = (1-t)Q_\alpha + tQ_\beta$$

$$W_p^p = \frac{1}{n} \sum_{i=1}^{n} |x_{(i)} - y_{(i)}|^p$$

Equal-weight samples: sort once, then match ranks. Cost: O(n log n).

# Wasserstein Barycenters

$$\bar{\alpha} \in \operatorname*{argmin}_{\alpha \in \mathcal{P}_2(\mathbb{R}^d)} \sum_{k=1}^{K} \lambda_k W_2^2(\alpha, \alpha_k)$$

$$\lambda_k \geq 0, \qquad \sum_k \lambda_k = 1$$

## An average defined by transport

Find one law that is close, in transport cost, to all of the input distributions.

**Point masses give the usual mean**

$$\bar{\alpha} = \delta_{\sum_k \lambda_k x_k} \quad \text{if } \alpha_k = \delta_{x_k}$$

**In 1D, the quantiles simply average**

$$\bar{Q} = \sum_k \lambda_k Q_k \qquad \text{in one dimension}$$

## Shape barycenters

Corners are inputs. Interior cells vary the weights.
Shown: entropy-regularized approximations.

# Gaussian Geometry: Location and Shape

$$W_2^2(\mathcal{N}(m_0,\Sigma_0),\mathcal{N}(m_1,\Sigma_1)) = \|m_0 - m_1\|^2 + \mathcal{B}^2(\Sigma_0,\Sigma_1)$$

$$\mathcal{B}^2(\Sigma_0,\Sigma_1) = \mathrm{tr}\Big(\Sigma_0 + \Sigma_1 - 2(\Sigma_0^{1/2}\Sigma_1\Sigma_0^{1/2})^{1/2}\Big)$$

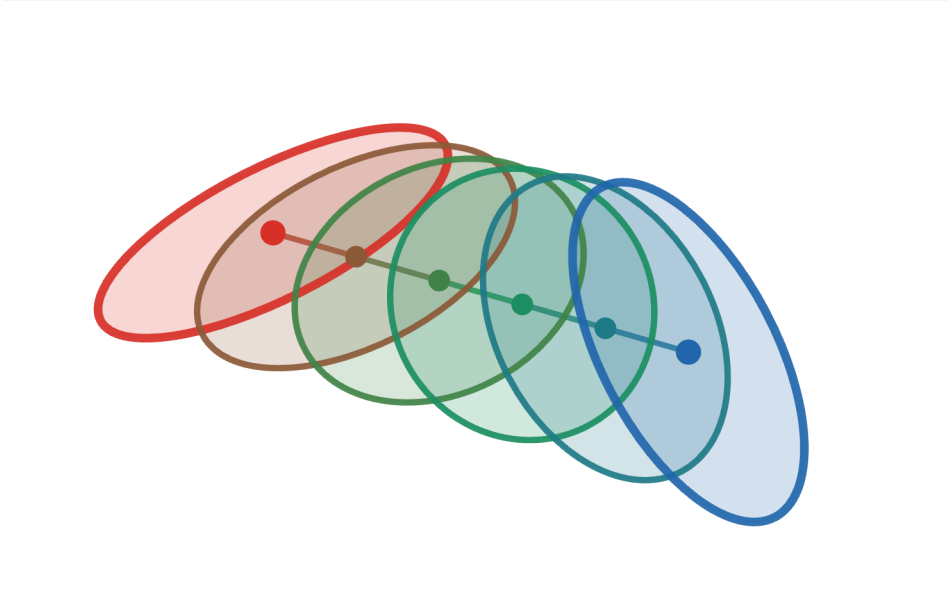
**FID!**

## In 1D: mean and standard deviation

$$W_2^2 = (m_0 - m_1)^2 + (\sigma_0 - \sigma_1)^2$$

$$m_t = (1-t)m_0 + tm_1, \qquad \sigma_t = (1-t)\sigma_0 + t\sigma_1$$

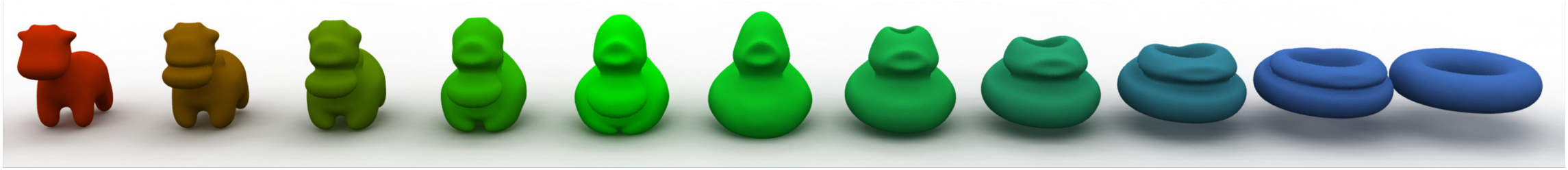
Standard deviations interpolate linearly.
Variances generally do not.



Means move linearly. Covariances follow a Bures path.

# 3D Shape Interpolation with Wasserstein Barycenters

## The same barycenter idea can interpolate 3D volumes

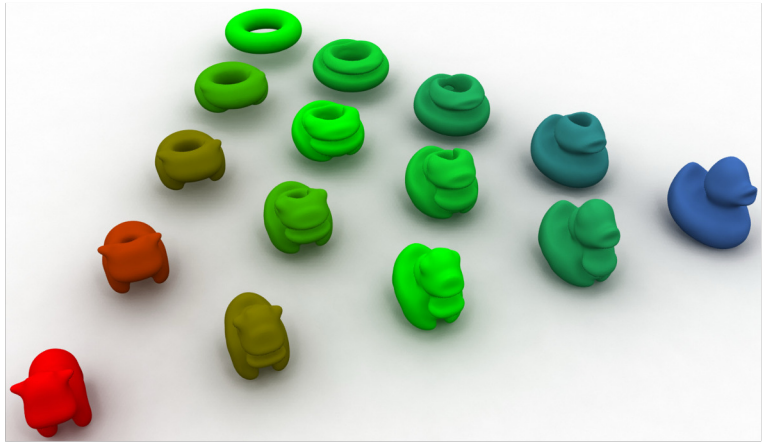


Each solid shape becomes a probability law over a grid of voxels.

## How OT helps

Barycenters move occupied volume. The resulting shapes can deform and change topology.

### Barycenters of three shapes



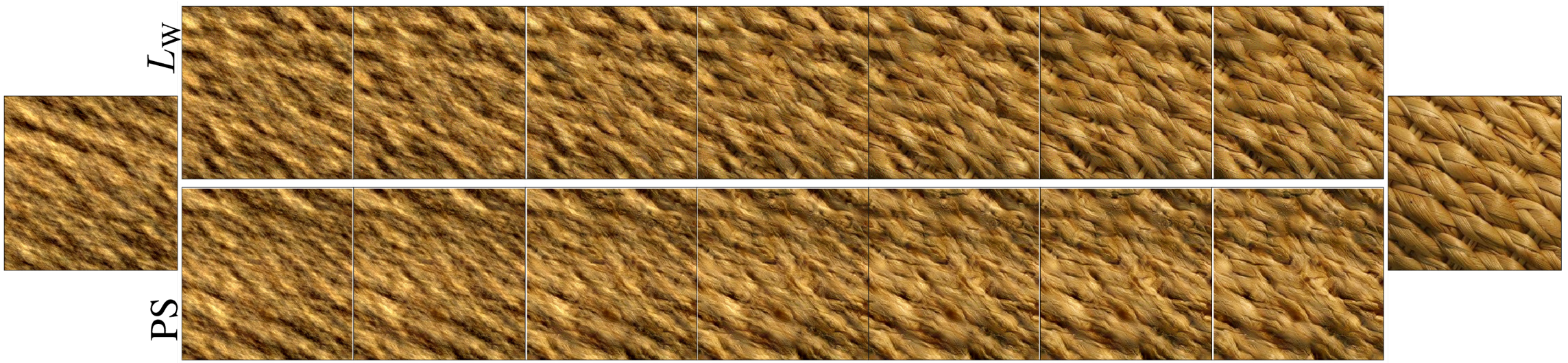
Solomon et al., Convolutional Wasserstein Distances, SIGGRAPH 2015. Figures 1 and 7.

# Neural Texture Interpolation

## Wasserstein geometry interpolates distributions of deep image features

Vacher et al., Texture Interpolation for Probing Visual Perception, NeurIPS 2020. Figure 5.

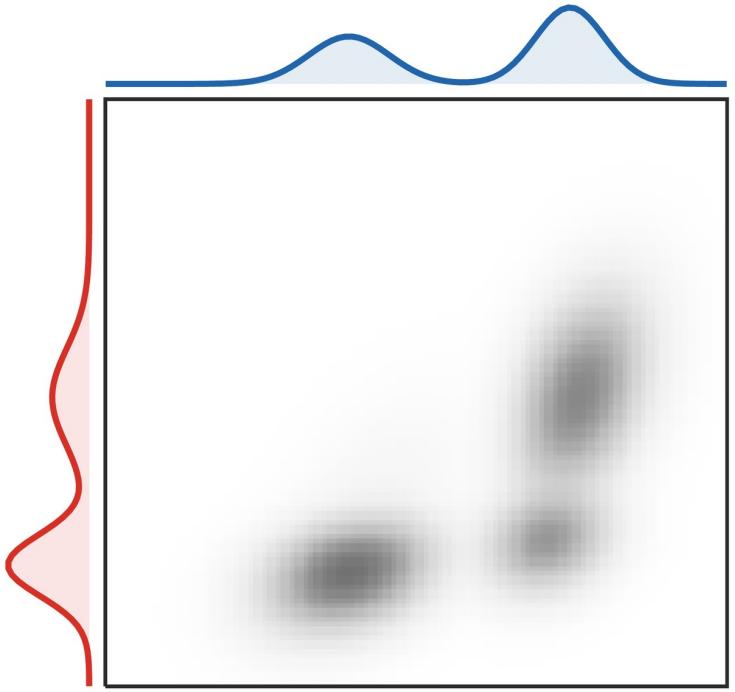
Wasserstein feature loss (top) and Portilla–Simoncelli baseline (bottom)

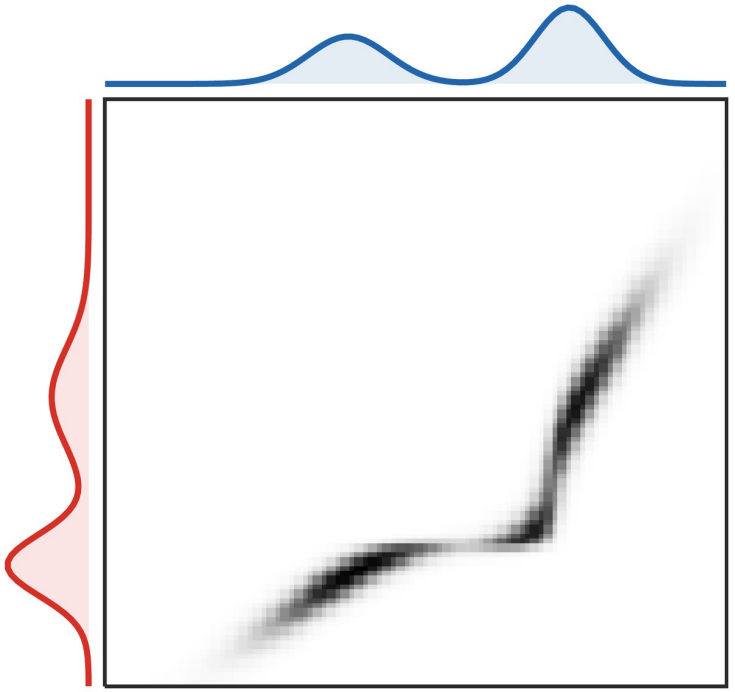



OT compares the means and covariances of CNN feature distributions.
An image is synthesized to match the interpolated statistics.

**Application: controlled texture transitions for visual-perception experiments**

# Computing OT





# Three Formulations of Optimal Transport

**Monge**

Optimize a map

$$\inf_{\substack{T:\mathcal{X}\to\mathcal{Y} \\ T_\sharp \mu = \nu}} \int_{\mathcal{X}} c\big(x, T(x)\big)\, d\mu(x)$$

**Kantorovich**

Optimize a coupling

$$\inf_{\pi \in \Pi(\mu,\nu)} \int_{\mathcal{X}\times\mathcal{Y}} c(x,y)\, d\pi(x,y)$$

**Kantorovich dual**

Optimize potentials

$$\sup_{\substack{f:\mathcal{X}\to\mathbb{R},\ g:\mathcal{Y}\to\mathbb{R} \\ f(x)+g(y)\le c(x,y)}} \left\{ \int_{\mathcal{X}} f(x)\, d\mu(x) + \int_{\mathcal{Y}} g(y)\, d\nu(y) \right\}$$

Adapted from Bunne & Cuturi, ICML 2023 tutorial, slide 59

# Kantorovich Potentials: Prices for Mass

$$\min_{P \geq 0} \langle C, P \rangle \quad \text{subject to} \quad P\mathbf{1}_m = a, \quad P^\mathsf{T}\mathbf{1}_n = b$$

Potentials reduce the number of unknowns and their storage.

$nm$ coupling entries $\longrightarrow$ $n + m$ potential values

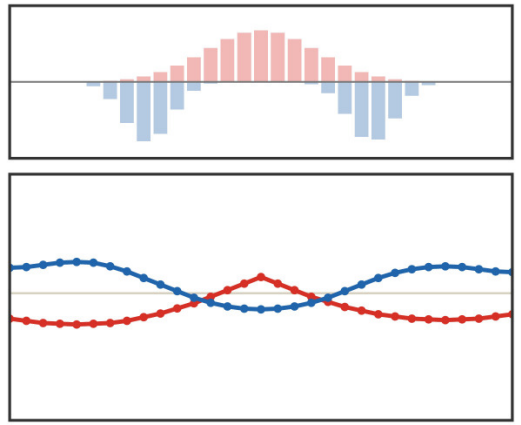
$$\mathcal{L}(P, f, g) = a^\mathsf{T} f + b^\mathsf{T} g + \sum_{i,j} (C_{ij} - f_i - g_j) P_{ij}$$

Any negative coefficient makes the infimum unbounded below.

$$\inf_{P \geq 0} \mathcal{L}(P, f, g) > -\infty \iff f_i + g_j \leq C_{ij}$$

$$\max_{f,g} \; a^\mathsf{T} f + b^\mathsf{T} g \quad \text{subject to} \quad f_i + g_j \leq C_{ij}$$

### Masses and their prices



Red: source / f. Blue: target / g.

### Certify an optimal plan

$$P^\star_{ij} > 0 \implies f^\star_i + g^\star_j = C_{ij}$$

Used routes saturate the price bound.

### Measure sensitivity

Potentials are subgradients of the optimal cost with respect to mass.

### Connect to Sinkhorn

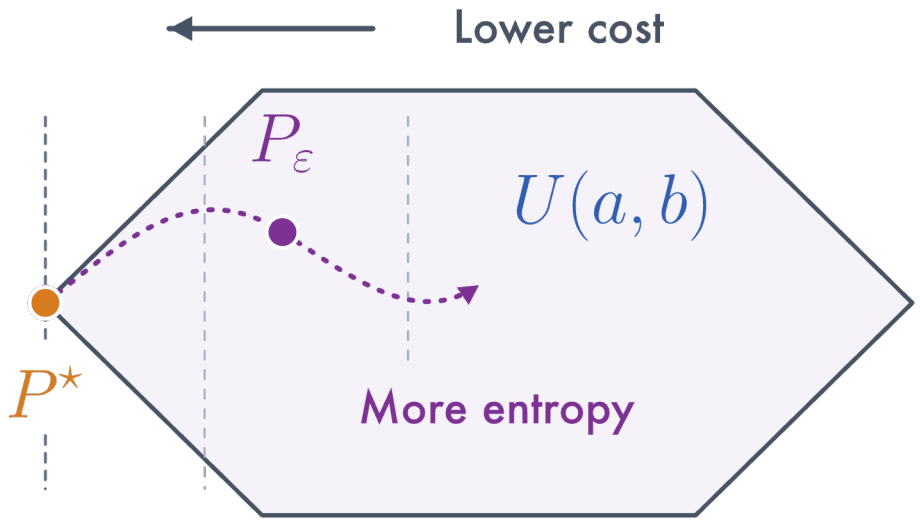
With entropy, exponentiating the potentials gives scaling factors.

# Entropy Makes Discrete OT Practical

$$P_\varepsilon = \arg\min_{P \in U(a,b)} \langle C, P \rangle - \varepsilon H(P), \qquad \varepsilon > 0$$

$$U(a,b) = \{P \geq 0 : \; P\mathbf{1}_m = a, \; P^\top \mathbf{1}_n = b\}, \qquad C_{ij} = c(x_i, y_j)$$

## Birkhoff Polytope



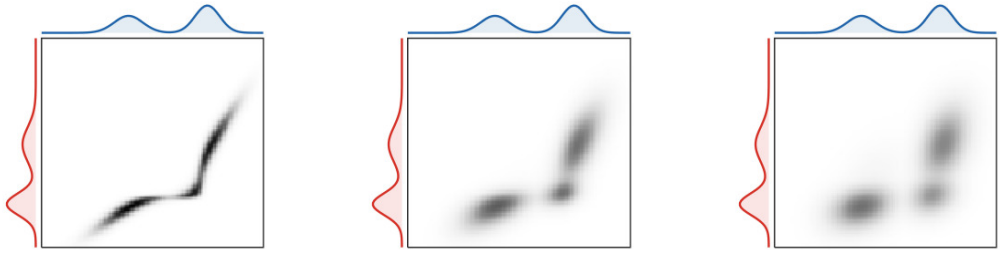

Every point is a coupling with the same marginals.

## Entropy of the coupling

$$H(P) = -\sum_{i,j} P_{ij} \log P_{ij}$$

Entropy measures uncertainty in the coupling table.
Rewarding it spreads mass across more pairs.

$\varepsilon = 0.01$ $\varepsilon = 0.085$ $\varepsilon = 0.24$



**Unique entropic plan** **Gradients for learning** **GPU-friendly updates**

Entropy introduces bias. More regularization spreads mass more widely.

# Sinkhorn: Alternate Row and Column Scaling

$$P_\varepsilon = \mathrm{diag}(u)\, K\, \mathrm{diag}(v), \qquad K_{ij} = e^{-C_{ij}/\varepsilon}$$

$$\underbrace{u_i = e^{f_i/\varepsilon}, \quad v_j = e^{g_j/\varepsilon}}$$

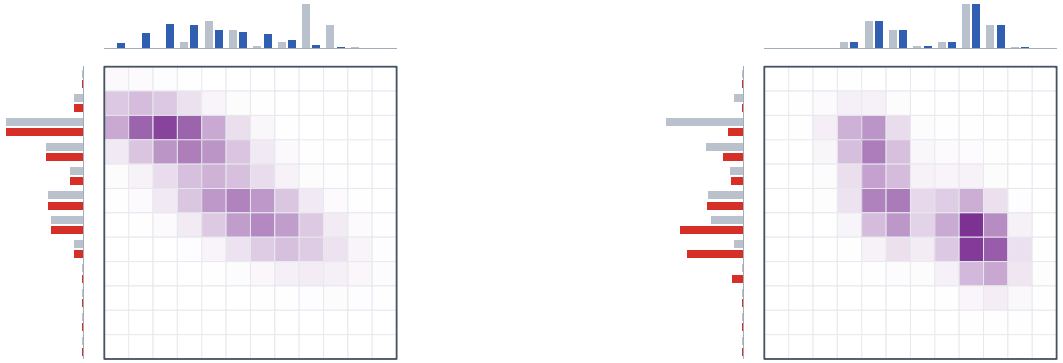
entropic dual potentials become scaling factors

$$u \leftarrow \frac{a}{Kv}$$

**Match source masses**

$$v \leftarrow \frac{b}{K^\mathsf{T} u}$$

**Match target masses**



Each update fixes one marginal and can disturb the other.

Gray: required mass. Red/blue: current mass. Purple: coupling.

**n-by-n entropic OT**

$\mathcal{O}(n^2)$

per iteration

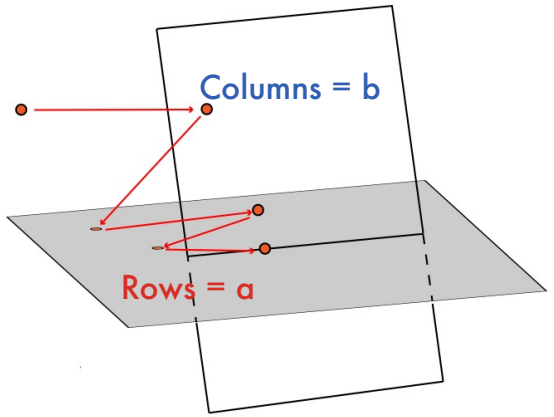
**GPU: 2 matvecs + elementwise divisions**

$\mathcal{O}(Tn^2)$

total for T iterations

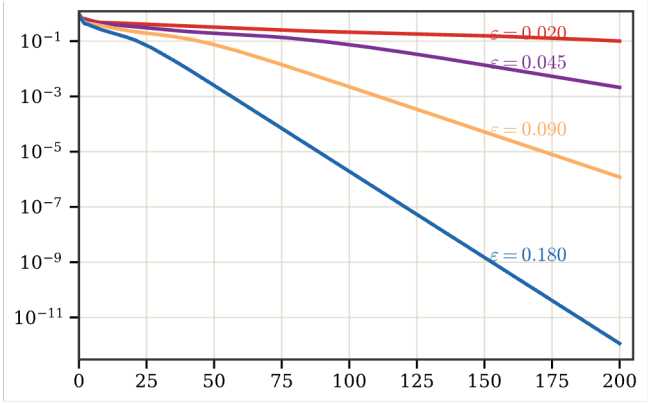
T depends on regularization and accuracy.

## Alternating KL projections




At the intersection, both marginals match

## Convergence of the marginal error




Half-steps (x), log error (y). Small epsilon is slower.

# From Dual Potentials to Optimal Maps

## Gangbo–McCann: the optimal coupling is induced by a map

Compact-support setting. The source has a density.

$$c(x,y) = h(x-y), \qquad h \in C^2(\mathbb{R}^d) \text{ strictly convex}, \qquad \mu \ll \mathcal{L}^d$$

### c-transform: one unknown function

$$g(y) = f^c(y) := \inf_x \{c(x,y) - f(x)\}$$

$$\sup_f \left\{ \int f \, d\mu + \int f^c \, d\nu \right\}$$

Optimize one potential. Feasibility is automatic.
The transform still needs an inner minimization.

### Quadratic cost: Brenier

$$c(x,y) = \tfrac{1}{2}\|x-y\|^2,$$

$$\phi(x) = \tfrac{1}{2}\|x\|^2 - f(x) \quad \text{is convex}$$

It is enough to search over convex potentials whose gradients match the target.

$$(\nabla\phi)_\sharp \mu = \nu$$

$$T^\star(x) = x - (\nabla h)^{-1}(\nabla f(x)) \qquad T^\star(x) = x - \nabla f(x) = \nabla\phi(x)$$

**Monge and Kantorovich coincide: same optimal cost, no mass splitting.**

$$\pi^\star = (\mathrm{Id}, T^\star)_\sharp \mu$$

Gangbo & McCann, Acta Mathematica 1996

Continue to Brenier's theorem

# Brenier's Theorem: A Convex Potential

## For quadratic cost, a source density gives a unique optimal map

$$\mu, \nu \in \mathcal{P}_2(\mathbb{R}^d), \qquad \mu \ll \mathcal{L}^d$$

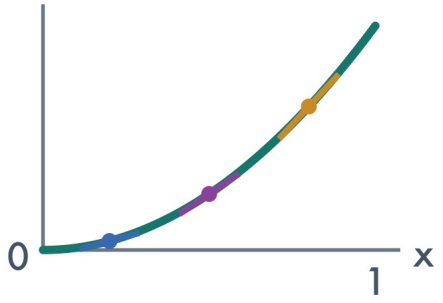
$$T^* = \nabla \phi, \qquad \phi \text{ convex}, \qquad T^*_{\#}\mu = \nu$$

$$\mu = \mathcal{U}[0,1], \qquad \nu = \mathcal{U}[0,2]$$

$$J(R) = \frac{1}{2}\mathbb{E}_{X\sim\mu}|X - R(X)|^2$$

### Convex potential

$$\phi(x) = x^2 \qquad T(x) = \phi'(x) = 2x$$

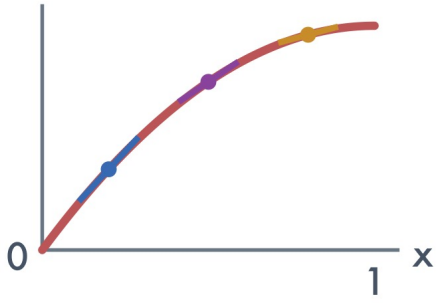



Slopes increase

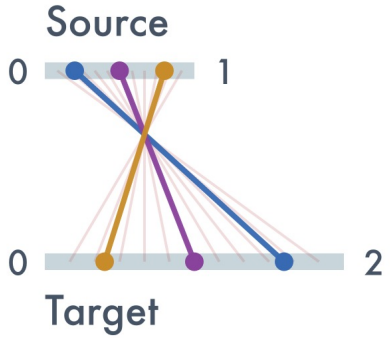
$$J(T) = \frac{1}{6}$$

### Concave potential

$$\psi(x) = 2x - x^2 \qquad S(x) = \psi'(x) = 2 - 2x$$




Slopes decrease



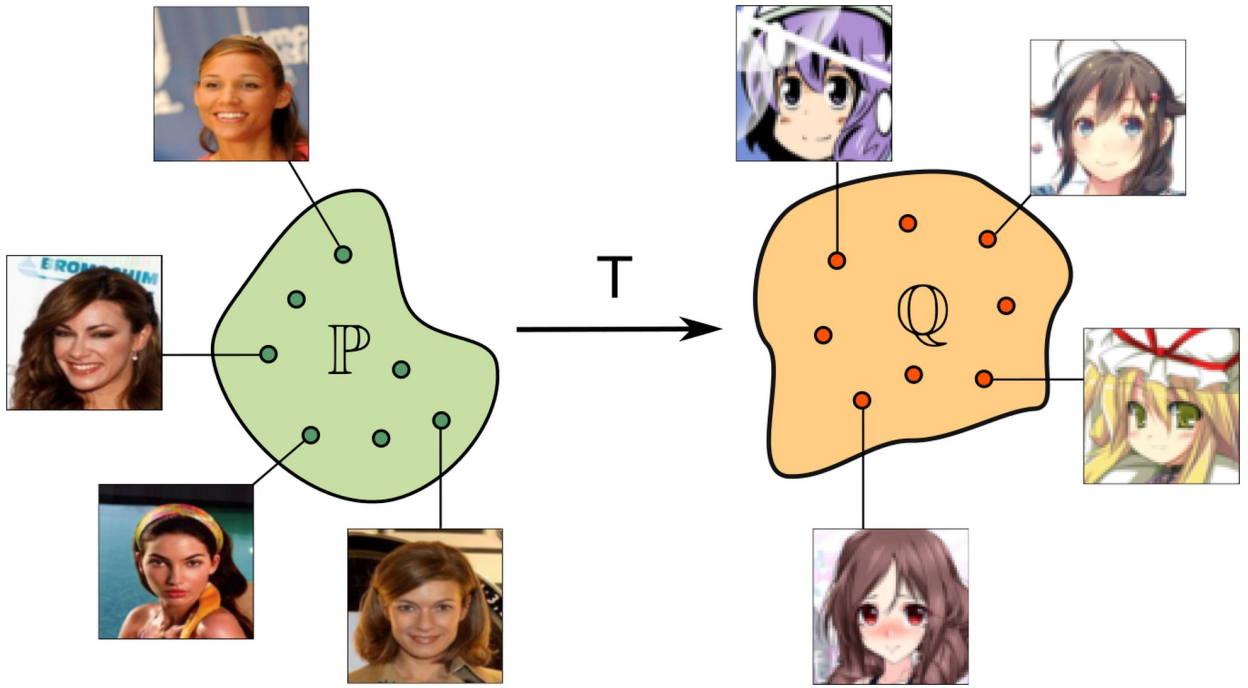

$$J(S) = \frac{1}{2}$$

In 1D, both maps reach the same target. Increasing slopes give the minimum cost.

Brenier (1991). Peyré, OT4ML, Monge chapter. Exact 1D example illustrated for this lecture.

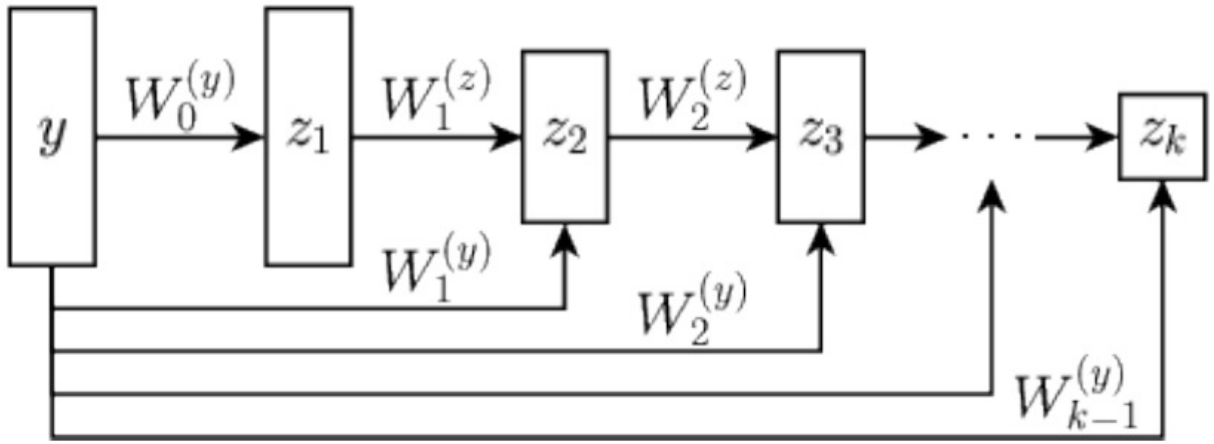
Back: dual-to-map recovery

# Generative Transport



# Input Convex Neural Networks

## Amos, Xu and Kolter, ICML 2017




### Convex in the input

$$W_\ell \geq 0 \quad \text{entrywise}$$

Use convex, nondecreasing activations, such as ReLU or softplus.

Input skip weights can have either sign.

$$z_{\ell+1} = \sigma_\ell(W_\ell z_\ell + A_\ell x + b_\ell)\,, \qquad \phi_\theta(x) = z_L$$

$$x \longmapsto \phi_\theta(x) \longmapsto \nabla_x \phi_\theta(x) = T_\theta(x)$$

Convexity in x does not make training convex in the network weights.

Amos, Xu and Kolter, Input Convex Neural Networks, ICML 2017. Original architecture from Fig. 1.

# Neural Dual Solvers

## Amos (ICLR 2023): amortized conjugates and L-BFGS

Paper

$$\min_{\theta} \underbrace{\mathbb{E}_{x\sim\mu}[f_\theta(x)] + \mathbb{E}_{y\sim\nu}\left[\max_z \{\langle z, y\rangle - f_\theta(z)\}\right]}_{D(\theta)\,:\,\text{quadratic dual, with a convex ICNN } f_\theta}$$

## Inner loop: optimize locations

$$z^*(y) = \operatorname*{argmin}_z \underbrace{f_\theta(z) - \langle z, y\rangle}_{J_\theta(z;y)}$$

$$z_0 = A_\eta(y) \xrightarrow{\text{L-BFGS}} \hat{z}(y) \approx z^*(y)$$

Freeze the potential. Predict a start, then refine it.

## Outer loop: update the potential

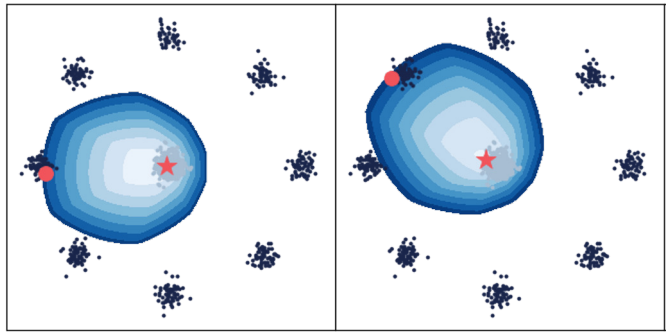
$$\nabla_\theta D \approx \mathbb{E}_\mu[\nabla_\theta f_\theta(x)] - \mathbb{E}_\nu[\nabla_\theta f_\theta(\hat{z}(y))]$$

Adam updates weights with refined locations held fixed.

Fit the initializer to refined solutions. Forward map:

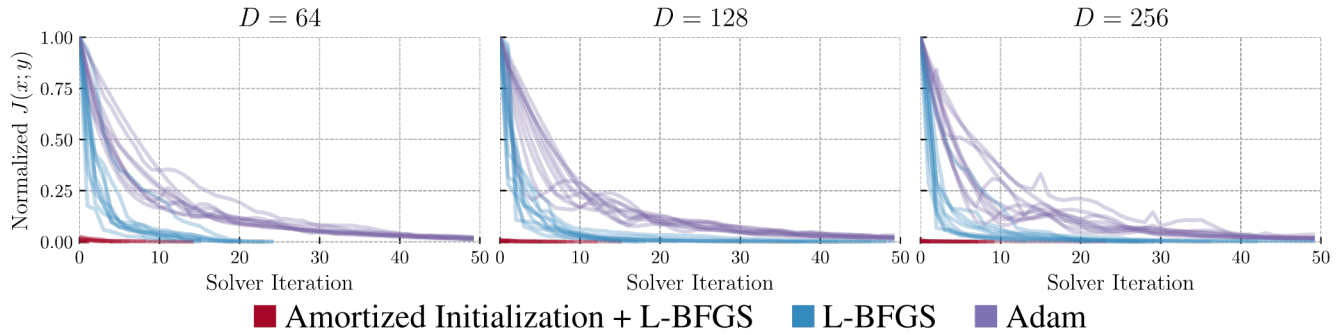
$$T_\theta(x) = \nabla_x f_\theta(x)$$

## A different landscape for each y



Dot = y. Star = minimizer.

## Inner convergence for a fixed ICNN

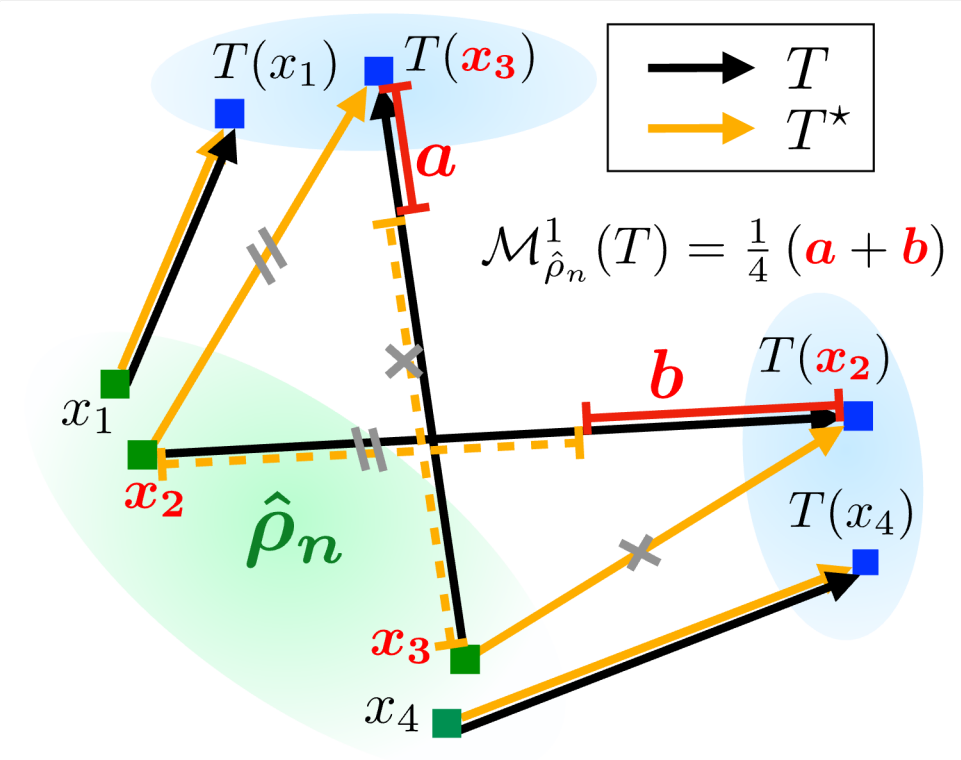



Figs. 6 and 2 from Amos (2023). Inner solves only. Outer training remains nonconvex.

# Monge Gap: Regularizing Neural Maps

## Distribution matching alone leaves the pairing undetermined

### Same endpoints, cheaper pairing




Black: learned map. Orange: optimal reassignment.
Euclidean distance. Red segments mark avoidable travel.

Uscidda & Cuturi, The Monge Gap, ICML 2023

### Measure the excess transport cost

$$\mathcal{M}^c_\mu(T) = \underbrace{\int c(x, T(x))\, d\mu(x)}_{\text{cost of the map}} - \underbrace{\mathrm{OT}_c(\mu, T_\sharp\mu)}_{\text{best cost for the same output}} \geq 0.$$

Zero gap: optimal transport to the map's own output.

### Train an unconstrained neural map

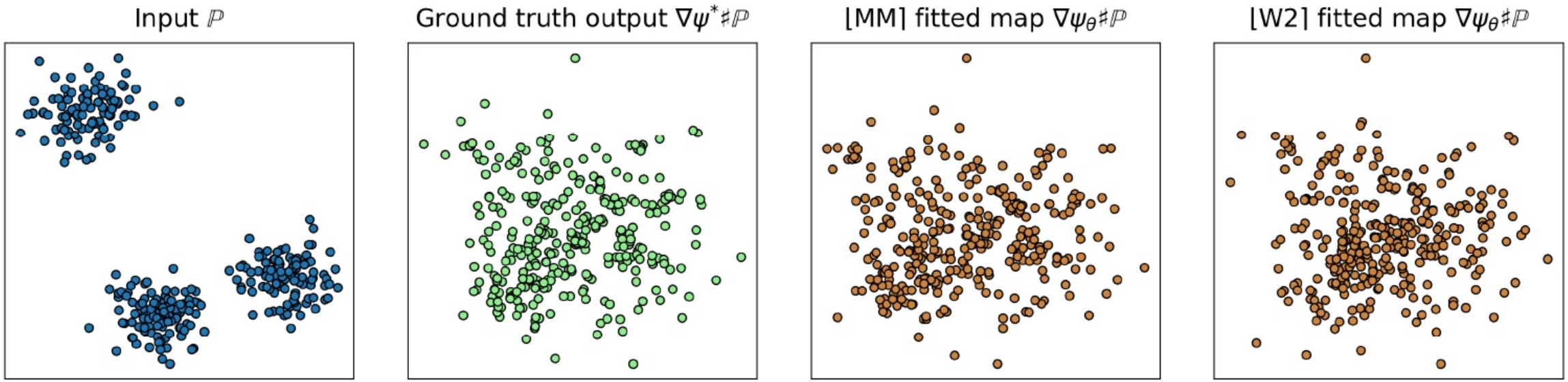
$$\min_\theta \underbrace{\Delta(T_{\theta\sharp}\mu, \nu)}_{\text{match target}} + \lambda \underbrace{\mathcal{M}^c_\mu(T_\theta)}_{\text{efficient map}}$$

Mini-batches + Sinkhorn estimate the regularizer.
Use Sinkhorn divergence to match the target.

OTT-JAX tutorial and code

# W2 Benchmark: Testing Neural OT Maps

## Construct a problem with a known optimal map




Matching the target's appearance does not certify the pointwise transport map.

## Generate the target with an ICNN gradient

$$\nu = (\nabla\phi_0)_{\#}\mu, \qquad T^* = \nabla\phi_0$$

## Measure error against the known map

$$\mathcal{L}_2^{\mathrm{UV}} = 100\,\frac{\mathbb{E}_\mu \|T_\theta(x) - T^*(x)\|^2}{\mathrm{Var}_\nu(Y)}$$

Korotin et al., Do Neural Optimal Transport Solvers Work? A Continuous Wasserstein-2 Benchmark, NeurIPS 2021.

# Neural Optimal Transport (NOT)

## A map and a dual potential

Strong cost, deterministic map

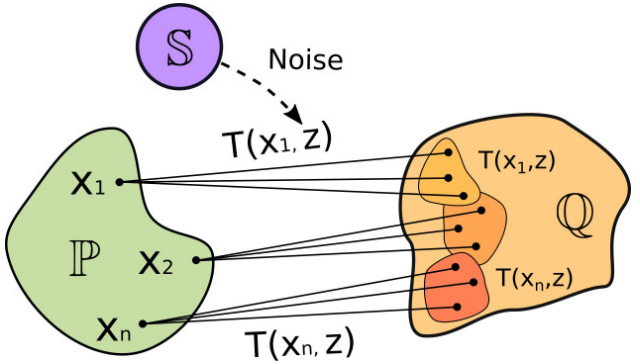
$$\sup_{g} \inf_{T} \quad \mathbb{E}_{x\sim\mu}[c(x, T(x)) - g(T(x))] + \mathbb{E}_{y\sim\nu}[g(y)]$$

Parameterize T and g with neural networks.
Alternate map and potential updates.

## Weak costs can model one-to-many plans

$$Y = T_\theta(X, Z), \qquad Z \sim \rho$$

## Unpaired image translation




Korotin, Selikhanovych and Burnaev, Neural Optimal Transport, ICLR 2023. Figures from the authors' repository.

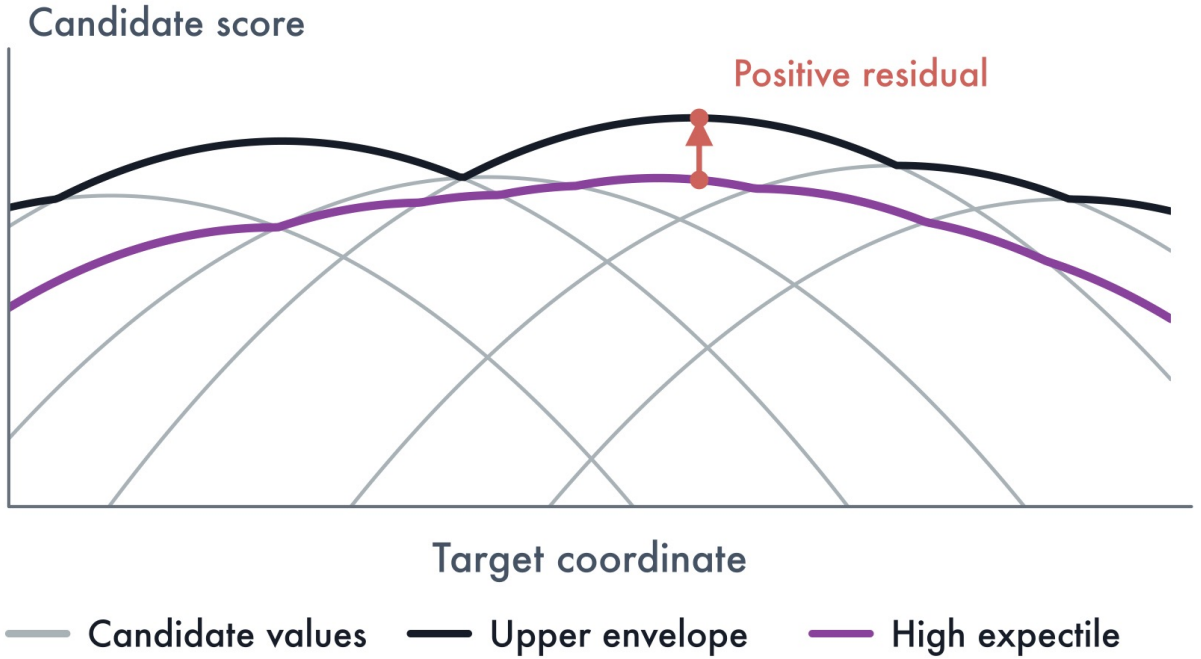
# ENOT: Approximate Conjugacy with Expectiles

## Fit conjugate potentials by asymmetric regression

ENOT project page

High-expectile regression replaces repeated inner conjugate optimization.

### Approximate the conjugate envelope




### Weight positive residuals

$$r = f(x) + g(y) - c(x, y), \qquad \mathcal{L}_\tau(r) = \left|\tau - \mathbf{1}_{\{r \le 0\}}\right| r^2$$

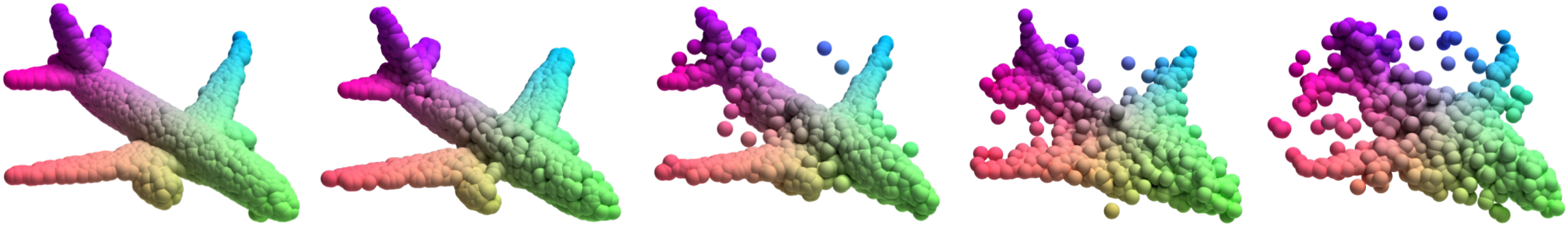
Buzun*, **Bobrin*** and Dylov, NeurIPS 2024 Spotlight.

# ENOT: Results in High Dimensions

## ShapeNet generation from Gaussian noise

ENOT project page

Target | **ENOT** | KernelOT | OT-FM | W2-OT



Each transported sample contains a full 3D point cloud.

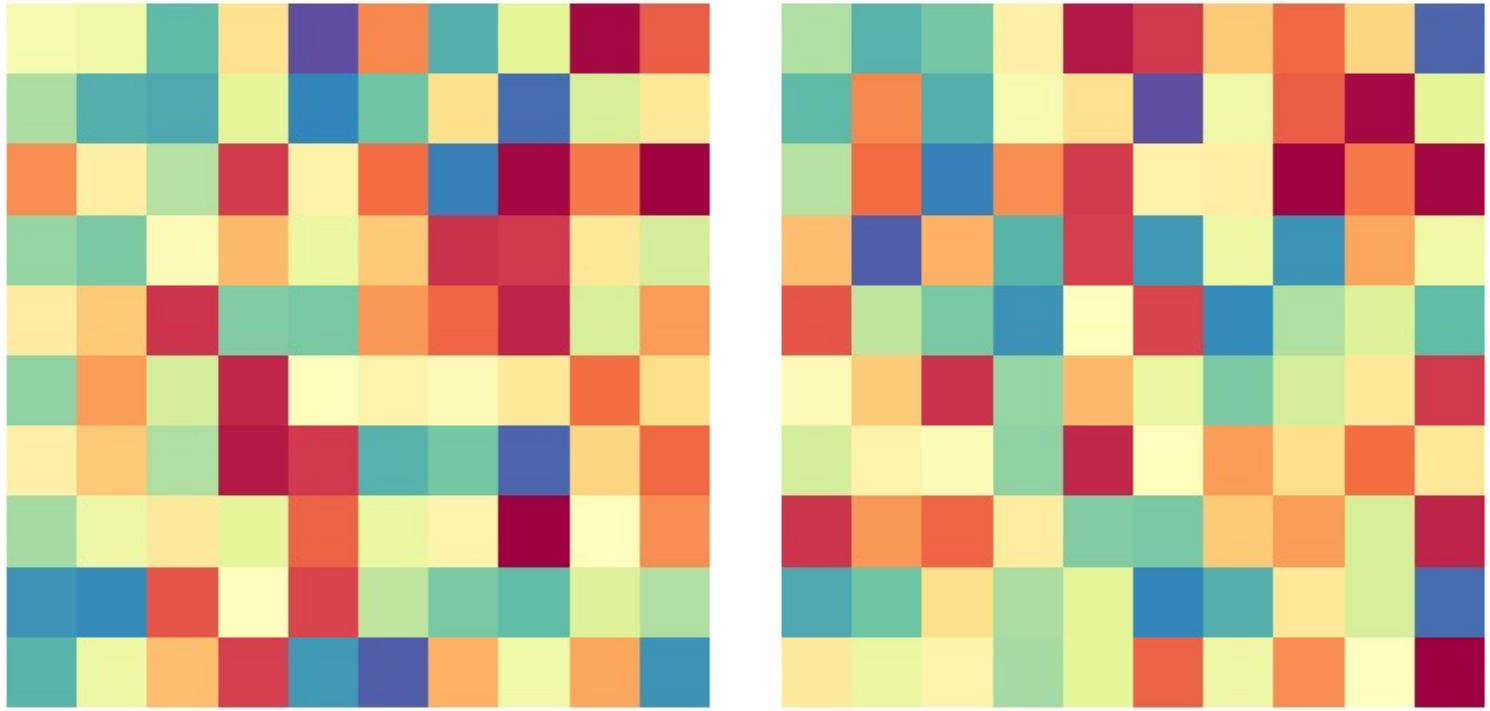
## Wasserstein-2 benchmark

Baseline: W2OT-Objective with L-BFGS

| Dim. | Error: baseline | Error: ENOT | Minutes: baseline | Minutes: ENOT |
|---|---|---|---|---|
| 64 | 2.08 ± 0.40 | **0.56 ± 0.03** | 397 | **16** |
| 128 | 0.67 ± 0.05 | **0.30 ± 0.01** | 571 | **21** |
| 256 | 0.59 ± 0.04 | **0.51 ± 0.02** | 1028 | **21** |

Error: unexplained variance (%), lower is better. Paper Tables 2 and 9. Airplanes: ENOT project page.

# Model Alignment



# Neurons as Discrete Measures

## A neuron becomes an atom described by its responses

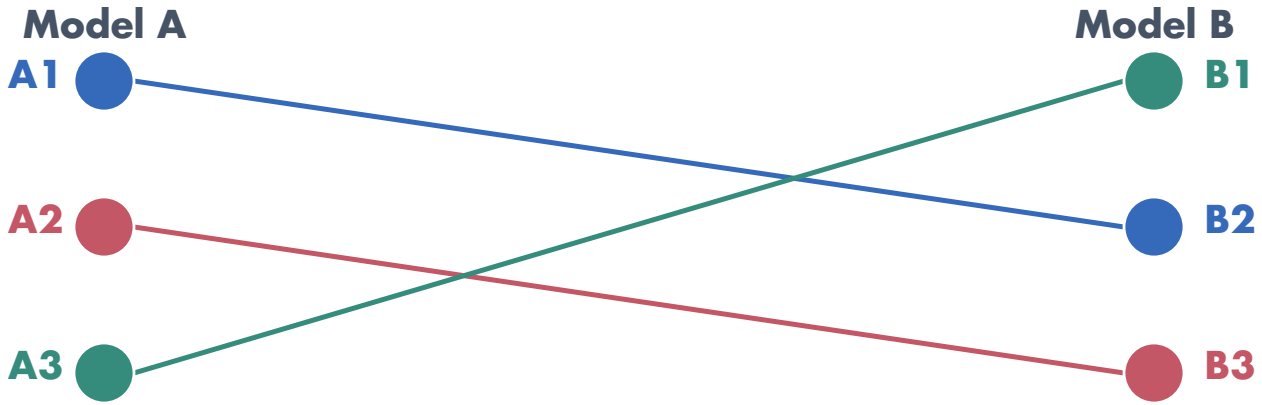
Paper

$$h_i = (a_i(x_1), \ldots, a_i(x_N)), \qquad C_{ij} = \|h_i^A - h_j^B\|^2$$

### Illustrative activation signatures

| Neuron | x1 | x2 | x3 | x4 |
|---|---|---|---|---|
| A1 | 1 | 1 | 0 | 0 |
| A2 | 0 | 1 | 1 | 0 |
| A3 | 0 | 0 | 1 | 1 |
| B1 | 0 | 0 | 1 | 1 |
| B2 | 1 | 1 | 0 | 0 |
| B3 | 0 | 1 | 1 | 0 |

### OT recovers the neuron correspondence




Zero-cost matches, each carrying mass 1/3

$$\mu_A = \sum_{i=1}^{n} a_i \, \delta_{h_i^A}, \qquad \mu_B = \sum_{j=1}^{m} b_j \, \delta_{h_j^B}$$

$$\Gamma^\star \in \operatorname*{argmin}_{\Gamma \geq 0} \langle C, \Gamma \rangle, \qquad \Gamma \mathbf{1} = a, \quad \Gamma^\top \mathbf{1} = b$$

Singh and Jaggi, Model Fusion via Optimal Transport, NeurIPS 2020. Toy activations are illustrative.

# Git Re-Basin: Linear Mode Connectivity

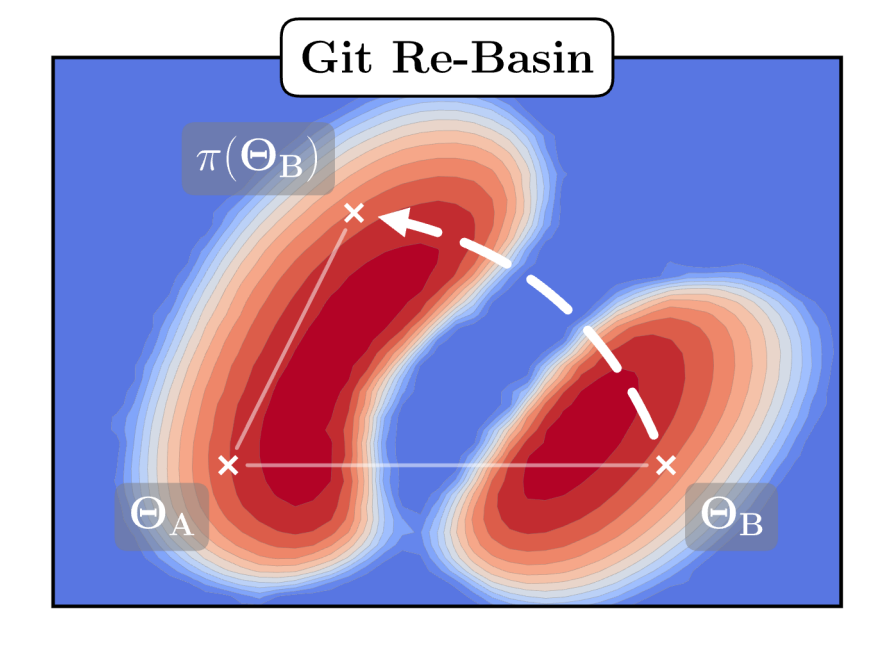
## A low-loss straight line makes weight averaging meaningful

Paper

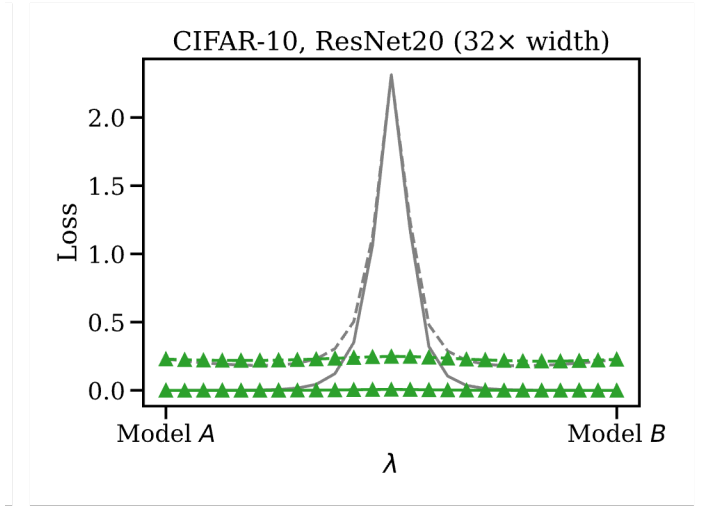
OT finds the neuron assignment. Re-Basin applies it consistently across layers.

$$P = n\Gamma^\star$$

**Relabeling moves B without changing its function**




**CIFAR-10, ResNet20 at 32× width**




Unaligned averaging

After neuron matching

$$\theta_\lambda = (1-\lambda)\theta_A + \lambda\,\pi_P(\theta_B), \qquad \lambda \in [0,1]$$

**Linear mode connectivity: the whole interpolation stays at low loss. The midpoint is a merged checkpoint.**

Ainsworth et al., Git Re-Basin, ICLR 2023, Figs. 1–2. Low barriers depend on the models and setting.

# Sinkhorn Re-Basin

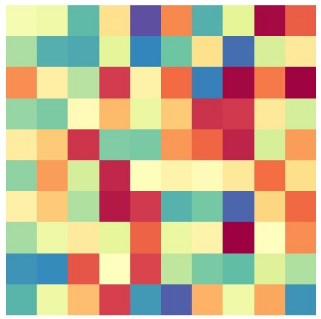
## Sinkhorn makes the neuron matching differentiable

Paper

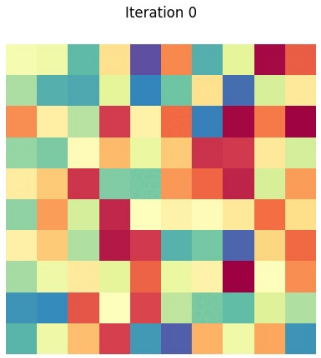
$$S_\tau(X) = \underset{S \in \mathcal{B}_n}{\operatorname{argmax}} \langle X, S \rangle + \tau H(S), \qquad S = \operatorname{diag}(u) e^{X/\tau} \operatorname{diag}(v)$$

$$\min_X \mathbb{E}_{\lambda \sim U[0,1]} \left[ \mathcal{L}\left( (1-\lambda)\theta_A + \lambda \pi_{S_\tau(X)}(\theta_B) \right) \right]$$

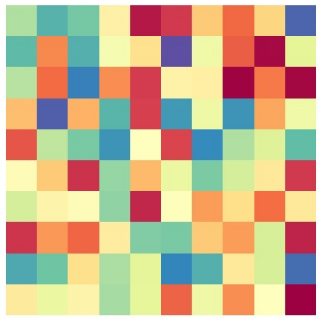
**Source weights**



**During alignment**




**Target weights**



## Implicit differentiation

Differentiate the optimality conditions of the entropic OT problem.

This avoids storing every Sinkhorn normalization step.

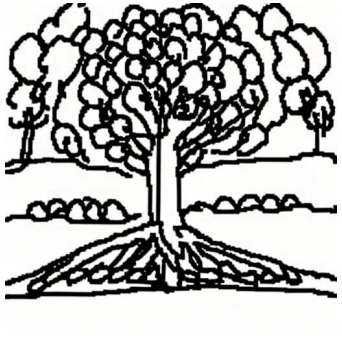
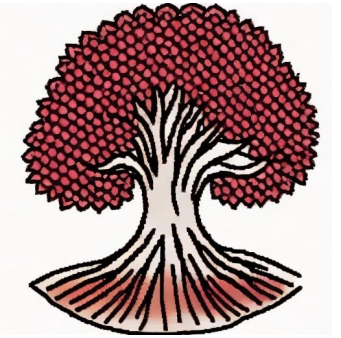
**Train with soft assignments. Project to a hard permutation for evaluation.**

Pena et al., Re-Basin via Implicit Sinkhorn Differentiation, CVPR 2023. Original repository animation.

# Model Merging Blends Learned Styles

## Fine-tuning learns styles that can be blended in weight space

Paper

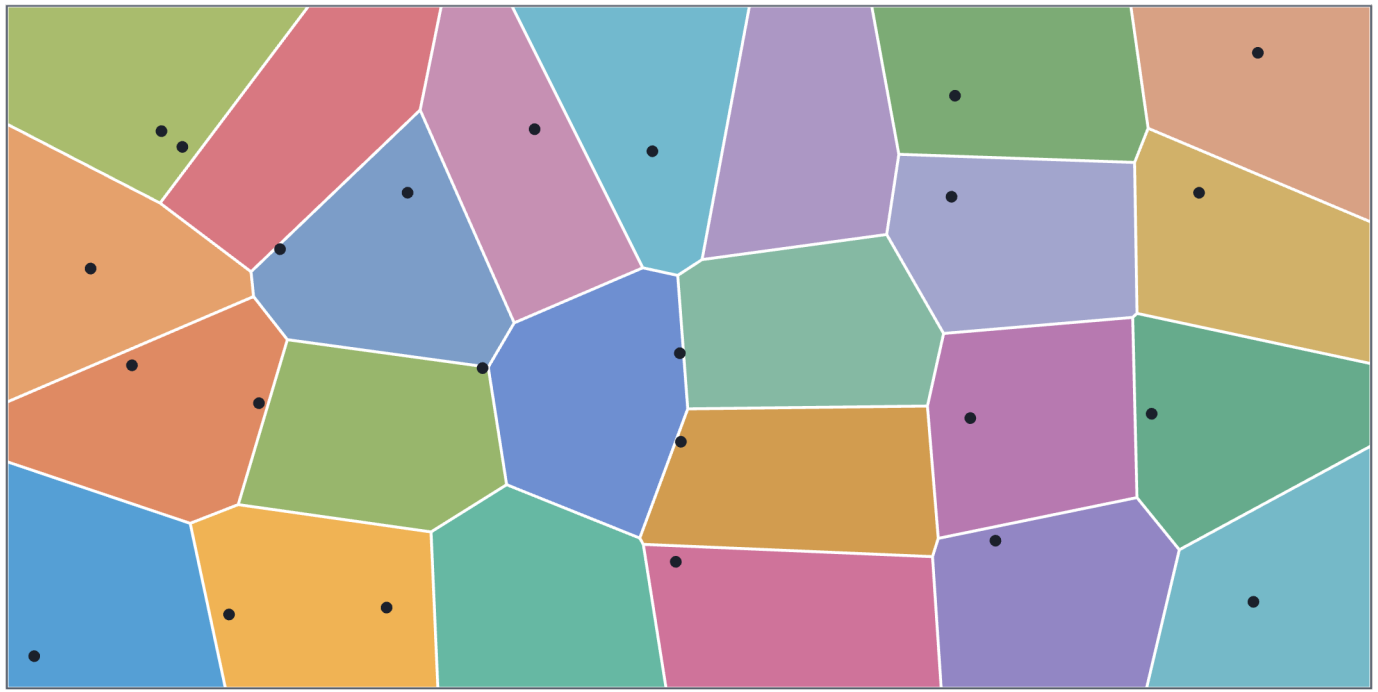
| | Pokémon fine-tune | | Sketch fine-tune | | Merged model |
|---|---|---|---|---|---|
| **Same prompt**<br>A tree surrounded by flowers | | + |  | = |  |
| **Same prompt**<br>A corgi with a bowtie and a party hat | | + | | = | |

$$\theta_{\text{soup}} = \tfrac{1}{2}\theta_{\text{Pokemon}} + \tfrac{1}{2}\theta_{\text{Sketch}}$$

**Diffusion Soup uses weight averaging. OT alignment helps when neuron coordinates differ.**

Biggs et al., Diffusion Soup, ECCV 2024, Fig. 5. Same base checkpoint, one model at inference.

# OT in Fluid Dynamics

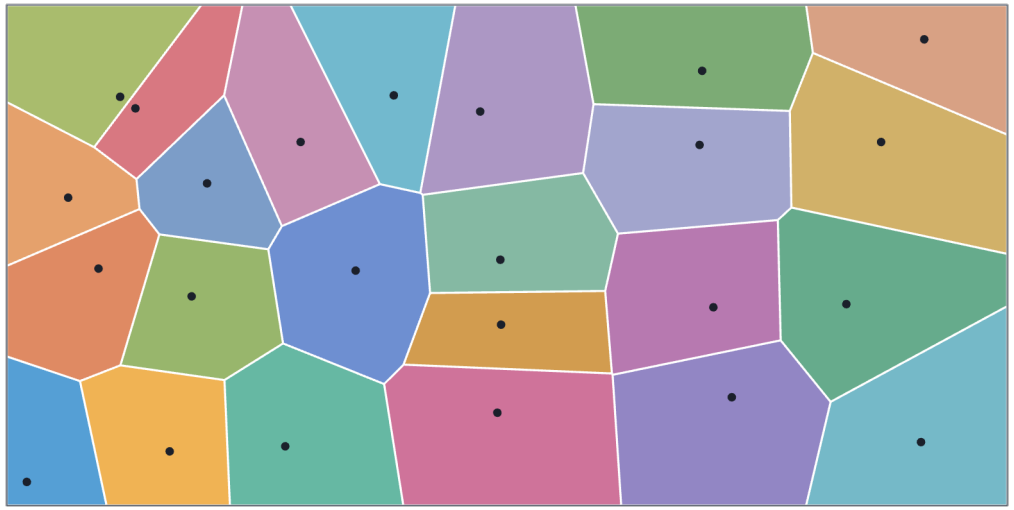


Cells change shape. Their volumes stay prescribed.

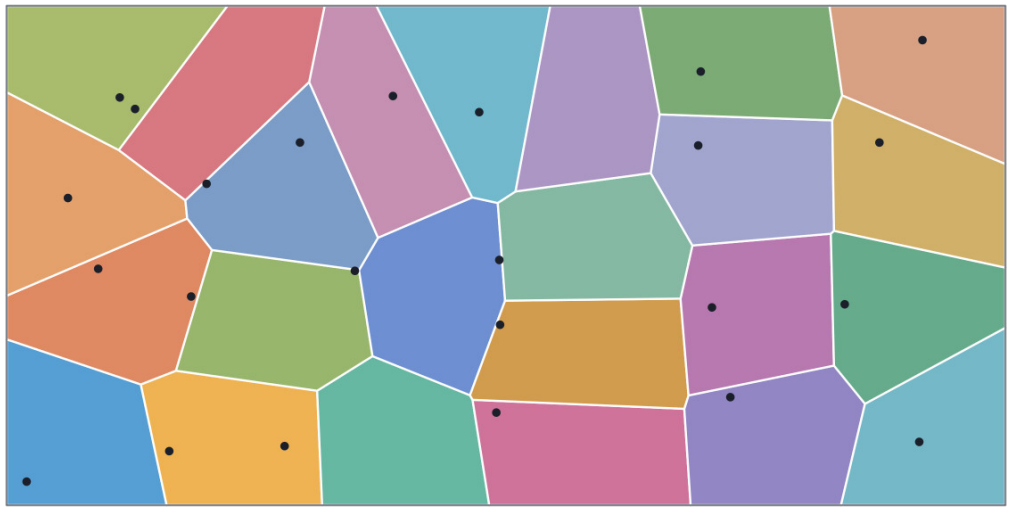
# Voronoi to Laguerre: Prescribed Volumes

## OT controls cell volumes by adjusting the dual weights

### Voronoi: geometry sets the volumes



### Laguerre: OT prescribes the volumes



Same sites and colours. Only the weights change. Sites can lie outside their Laguerre cells.

$$L_i(w) = \left\{ x \in \Omega : \ \|x - q_i\|^2 - w_i \leq \|x - q_j\|^2 - w_j, \ \forall j \right\}$$

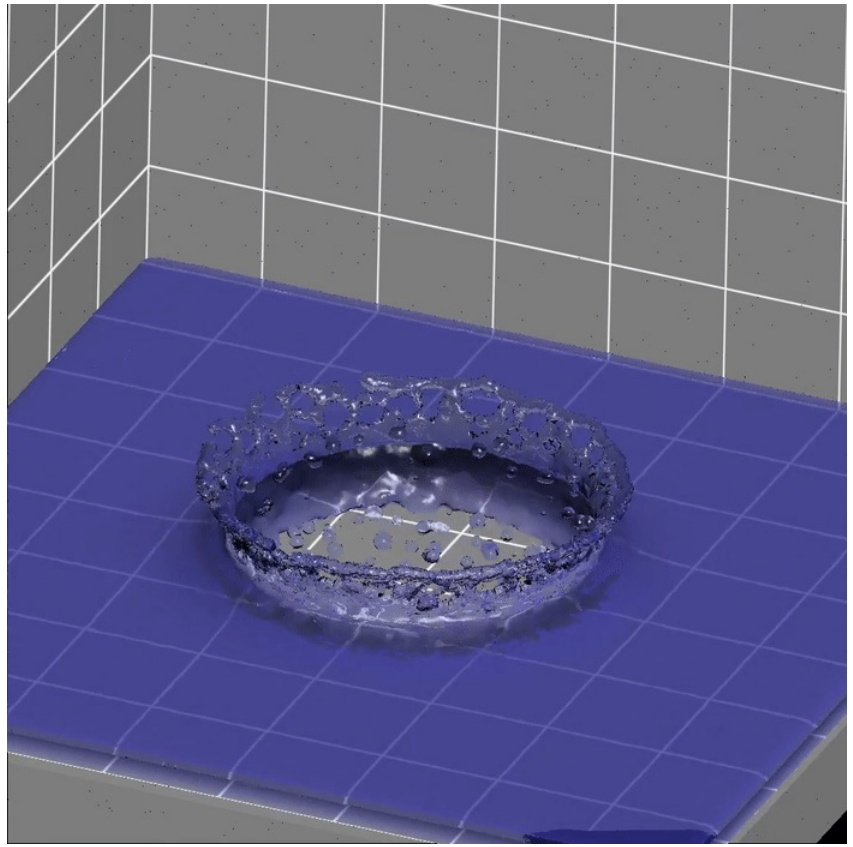
$$\nabla_i G(w) = V_i - |L_i(w)| = 0 \quad \Longrightarrow \quad |L_i(w^\star)| = V_i$$

Plateau-Holleville & Lévy, OT for Fluid Simulation, slides 67–76

# From OT Cells to Fluid Motion

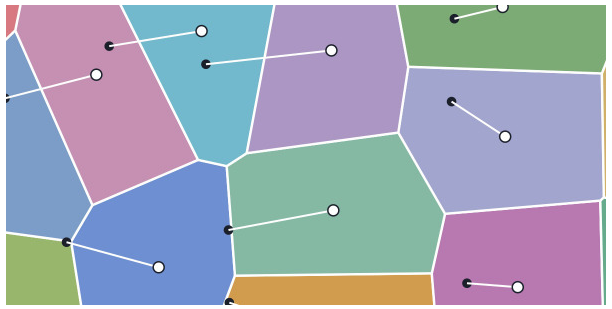
## Use the OT cells to drive incompressible motion

### 3D free-surface splash



Drop impact, crown formation and droplets.
Bruno Lévy, Graphite OT fluid simulation

### A related OT update: centroid springs



Dark points: particle sites.
White points: cell centroids.
The restoring force pulls sites toward their centroids.

$$c_i = \frac{1}{V_i} \int_{L_i(w^\star)} x \, dx$$

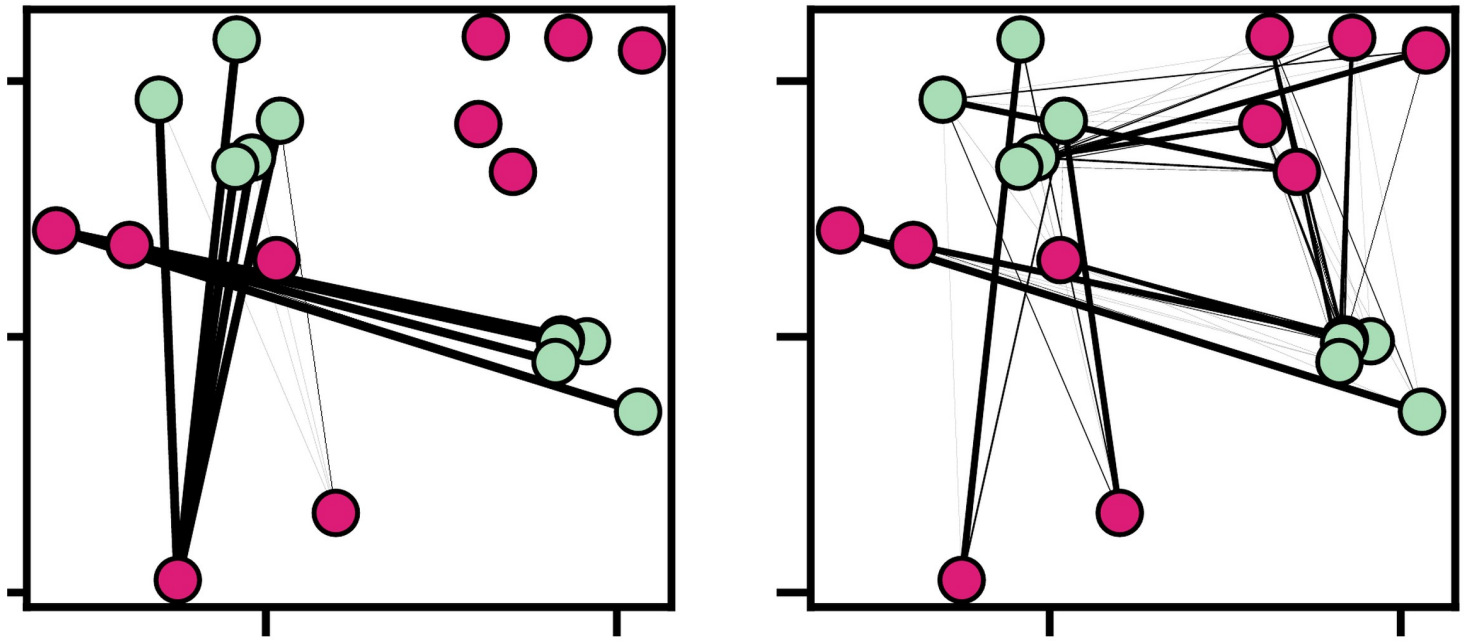
$$\ddot{q}_i = \frac{c_i - q_i}{\eta^2} + g$$

A spring toward the OT cell centroid, plus gravity.

$$E_\eta(q) = \frac{1}{2\eta^2} W_2^2 \left( \sum_i V_i \delta_{q_i}, \, dx|_\Omega \right)$$

Gallouët & Mérigot, FoCM (2018), Sections 3–5

# Mixing & Routing



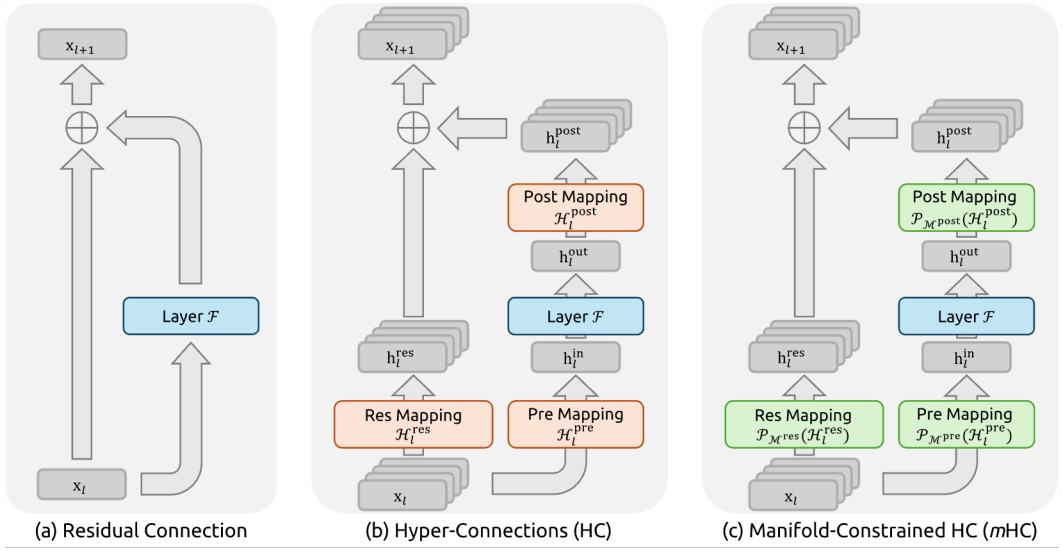
# mHC: Residual-Stream Computation

## A layer reads from the streams and writes its update back

Paper

$$X_{l+1} = H_l^{\text{res}} X_l + (H_l^{\text{post}})^\top F(H_l^{\text{pre}} X_l)$$

For each token: n feature streams. F is token attention or an MLP.




(a) Residual Connection (b) Hyper-Connections (HC) (c) Manifold-Constrained HC (*m*HC)

### Read into F

Combine the streams before the Transformer sublayer.

### Write the update back

Redistribute the sublayer output across the streams.

### Mix the residual paths

Sinkhorn constrains the learned n × n to be Doubly Stochastic !

$$X_L = \underbrace{H_{L-1}^{\text{res}} \cdots H_l^{\text{res}}}_{\text{many matrix multiplications}} X_l + \cdots$$

mHC in DeepSeek-V4 (2026)

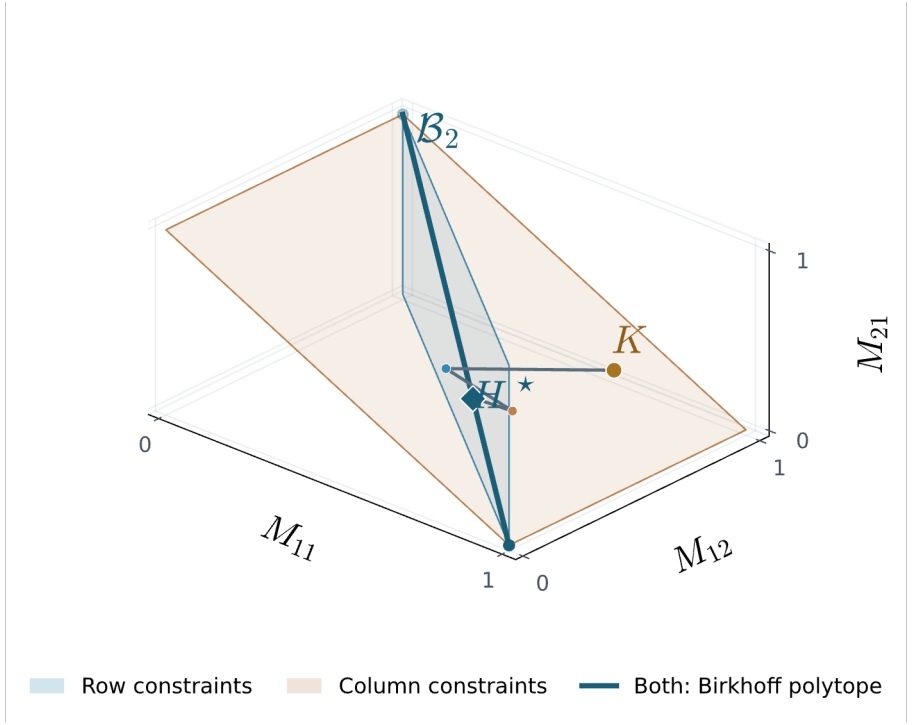
Related: Kimi AttnRes (2026), attention over depth

# mHC: Sinkhorn as an OT Projection

## Turn learned scores into balanced residual-stream mixing

Paper

### OT view: a KL projection onto the Birkhoff polytope




$$H^\star = \underset{H \in \mathcal{B}_n}{\operatorname{argmin}} \ \mathrm{KL}(H \,\|\, K), \quad K = \exp Z_l$$

2×2 illustration. The paper uses n = 4 streams.

### From learned scores to the mixer

$$M^{(0)} = \exp(Z_l) > 0$$

Input-dependent scores, exponentiated elementwise.

### 1. Normalize columns

$$C_{ij}^{(t)} = \frac{M_{ij}^{(t)}}{\Sigma_r M_{rj}^{(t)}}$$

### 2. Normalize rows

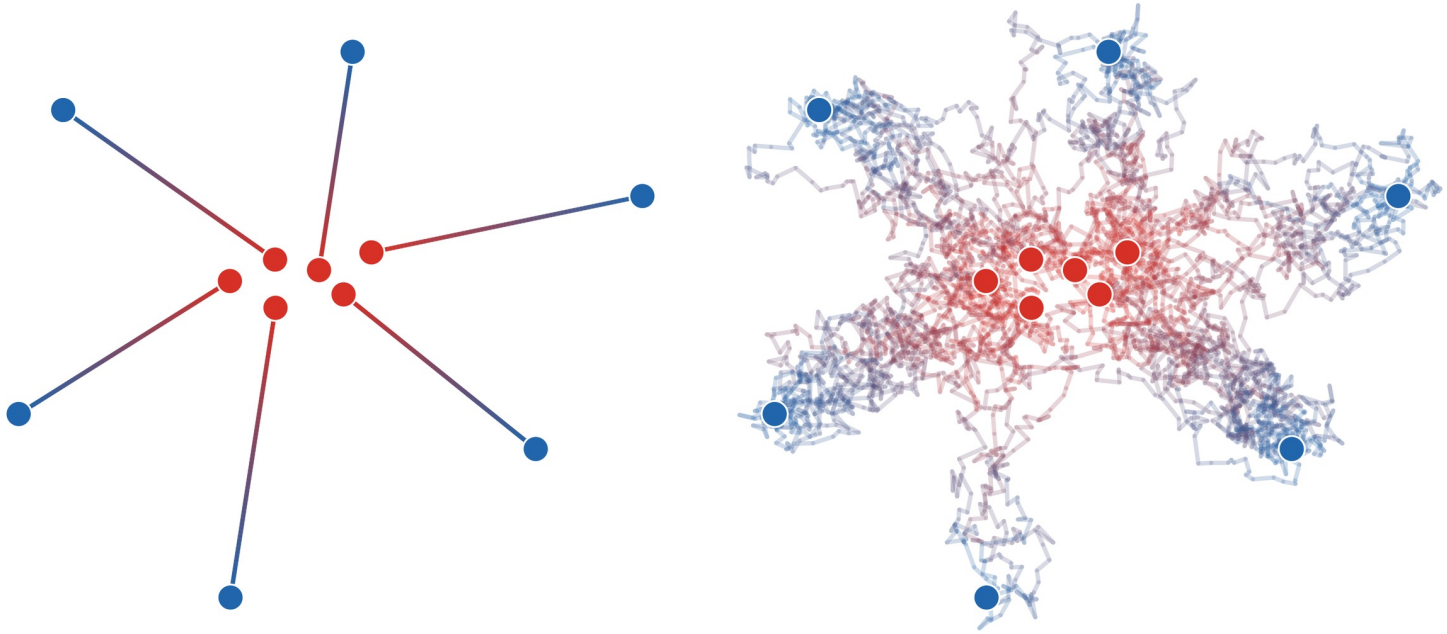
$$M_{ij}^{(t+1)} = \frac{C_{ij}^{(t)}}{\Sigma_s C_{is}^{(t)}}$$

$$H_l^{\mathrm{res}} = M^{(T)}, \qquad T = 20$$
$$H_l^{\mathrm{res}} \mathbf{1} \approx \mathbf{1}, \qquad (H_l^{\mathrm{res}})^\top \mathbf{1} \approx \mathbf{1}$$

Repeat both steps. Exact balance is a convergence limit; 20 iterations give the paper's practical approximation.

Xie et al., mHC, Eqs. 8–9. One iteration normalizes columns, then rows. T = 20 in the paper.

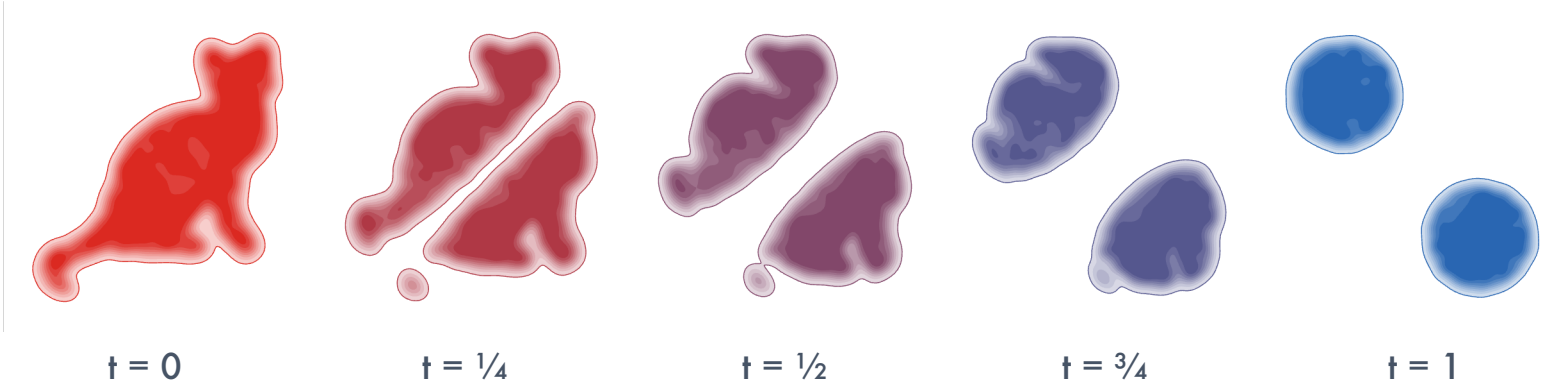
# Schrödinger Bridges / Dynamic OT



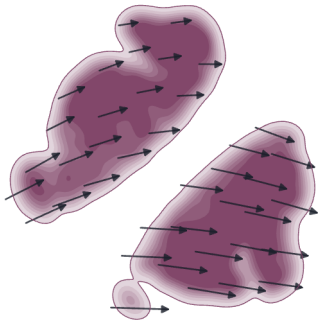
# Dynamic OT: Benamou–Brenier

## Move a density between two endpoints with the least kinetic energy

### A whole path of densities




### A velocity at each location



Arrows show the direction of motion.

$$W_2^2(\mu,\nu) = \inf_{(\rho_t, v_t)} \int_0^1 \int_{\mathbb{R}^d} \rho_t(x)\, \|v_t(x)\|^2 \,\mathrm{d}x\,\mathrm{d}t$$

**Mass conservation**

$$\partial_t \rho_t + \nabla \cdot (\rho_t v_t) = 0$$

**Fixed source and target**

$$\rho_0\,\mathrm{d}x = \mu, \qquad \rho_1\,\mathrm{d}x = \nu$$

**Same quadratic OT cost, expressed as the energy of a moving density.**

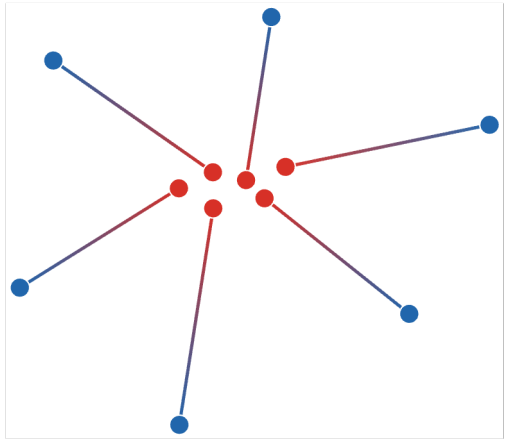
Next: Schrödinger bridges add a diffusion prior to the path description.

Benamou & Brenier (2000). Figures adapted from G. Peyré, OT4ML: Dynamic Optimal Transport.

# Why Schrödinger Bridges?

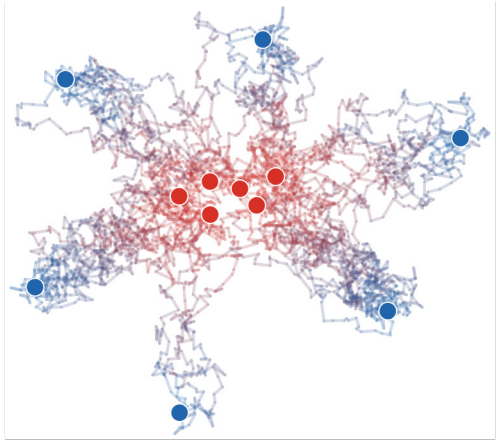
## Two snapshots leave the trajectories unknown

### Classical OT



Minimize displacement cost. Straight quadratic-cost paths.

### Schrödinger bridge



Stay close to a diffusion prior. Random paths, fixed endpoints.

Inputs: $(x_i)_i, (y_i)_j$.

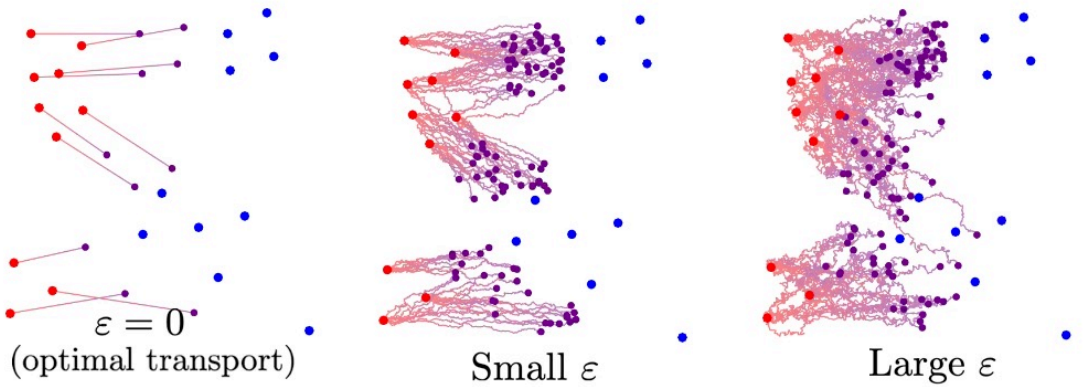
Temperature $\varepsilon$.

Entropic optimal transport:

$$\min_{P \in \mathbb{R}_+^{n \times m}} \left\{ \sum_{i,j} P_{i,j} \|x_i - y_j\|^2 + \varepsilon P_{i,j} \log(P_{i,j}) \; ; \; P \mathbb{1}_m = \mathbb{1}_n, P^\top \mathbb{1}_n = \mathbb{1}_m \right\}$$

Schrödinger interpolation at time $0 \leqslant t \leqslant 1$:

$$\sum_{i,j} P_{i,j} \, \mathcal{N}((1-t)x_i + t y_j, \varepsilon t(1-t))$$




Animation: G. Peyré, Mathematical Nexus

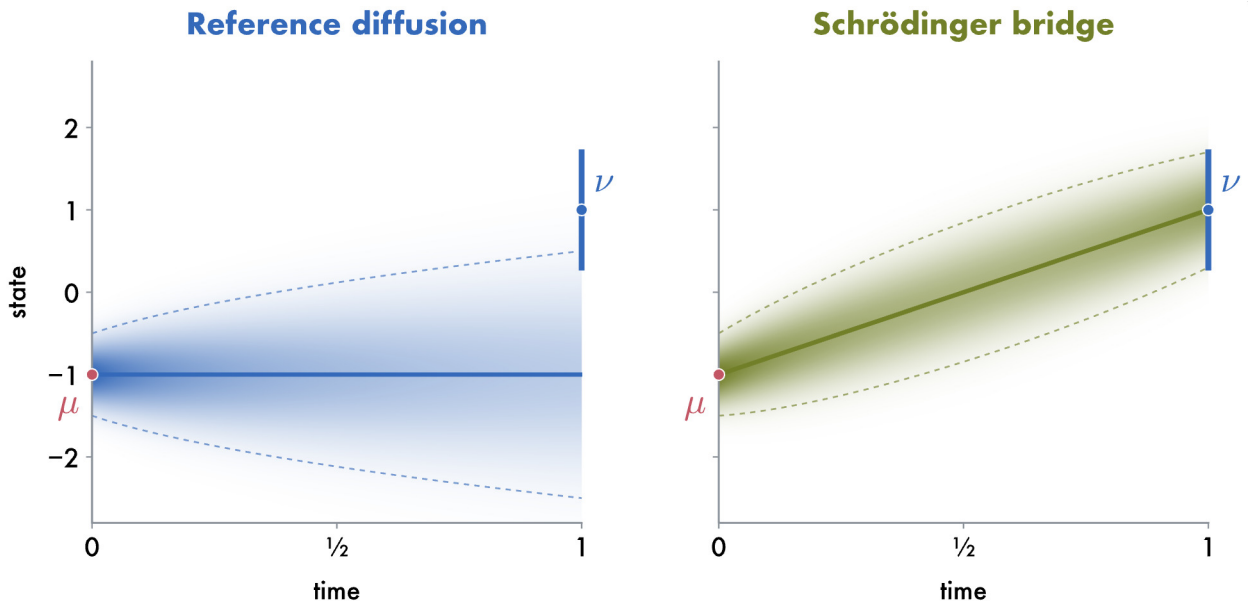
**Useful when uncertainty matters: cell trajectories and image restoration.**

C. Léonard (2014). Applications: Liu et al. (2023), Tong et al. (2024).

# Schrödinger's Problem on Path Space

## A whole law of trajectories, constrained at both ends

Source




Color = density. Solid = mean. Dashes = ±2 standard deviations.

**The bridge steers the prior toward the target.**

### Closest path law to a prior

$$P^\star = \underset{P_0=\mu,\, P_1=\nu}{\arg\min}\ \mathrm{KL}(P \,\|\, R^\varepsilon)$$

Brownian prior, started from the source law

### Change the drift, keep the diffusion

$$\mathrm{d}X_t = u_t(X_t)\,\mathrm{d}t + \sqrt{\varepsilon}\,\mathrm{d}W_t$$

$$\varepsilon\,\mathrm{KL}(P\|R^\varepsilon) = \mathbb{E}_P \textstyle\int_0^1 \frac{1}{2}\|u_t(X_t)\|^2\,\mathrm{d}t$$

Least control energy to reach the target law

C. Léonard (2014). Exact Gaussian example: Bunne et al., AISTATS 2023.

# Entropic OT Determines the Endpoints

## An entropic coupling, filled with conditional Brownian bridges

Source

$$\pi_\varepsilon^\star = \underset{\pi \in \Pi(\mu,\nu)}{\arg\min} \int \tfrac{1}{2}\|x-y\|^2 \,\mathrm{d}\pi + \varepsilon\,\mathrm{KL}(\pi \| \mu \otimes \nu)$$

### 1 Pair endpoints with Sinkhorn

$$\pi_{ij}^\star = u_i \exp\left(-\frac{\|x_i - y_j\|^2}{2\varepsilon}\right) v_j$$

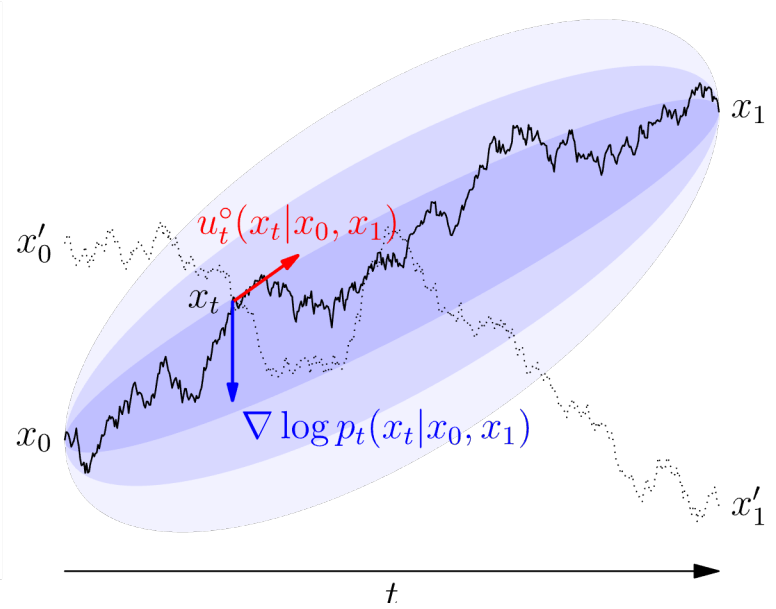
Scaling factors enforce the two marginals.

### 2 Sample a conditional Brownian bridge

$$X_t \mid x, y \sim \mathcal{N}(m_t, \varepsilon t(1-t) I)\,,$$

$$m_t = (1-t)x + ty$$

**Zero-noise limit: quadratic OT**




Conditional means and scores are known analytically.

G. Peyré, OT4ML. C. Léonard (2014). Figure: Tong et al., Simulation-Free Schrödinger Bridges, Fig. 1 (2024).

# Schrödinger Bridges in Practice

## Bridge models connect data distributions with useful structure

### Image restoration

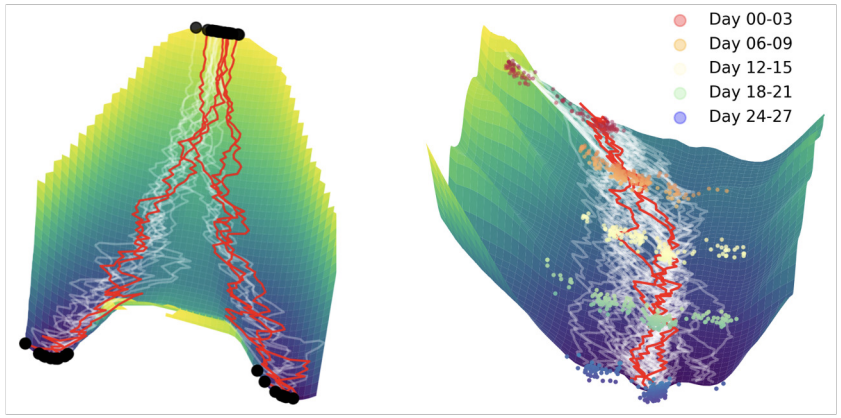
I²SB, ICML 2023

Input Bridge sample Reference

I²SB learns diffusion from degraded to clean images, using paired training examples.

### Cell dynamics from snapshots

SF²M, AISTATS 2024




Toy branching (left), embryoid-body data (right).

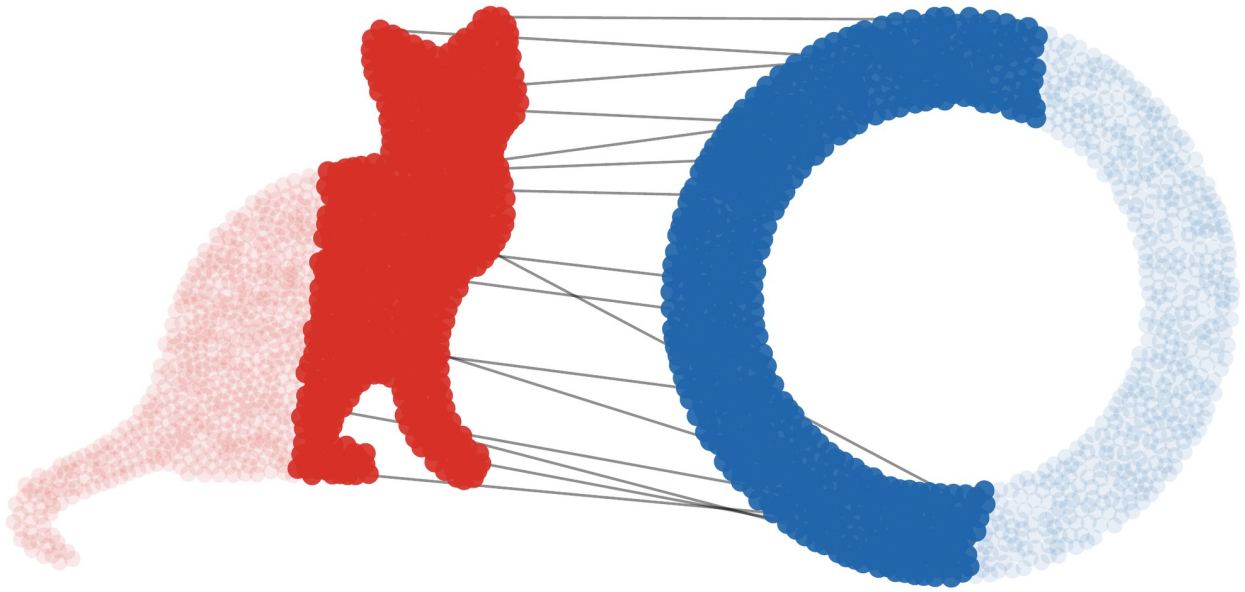
SF²M uses entropic OT pairing and score/flow regression to infer stochastic dynamics.

Inferred paths depend on the prior and cost.

### Two informative endpoint distributions

Further reading: Tilted Schrödinger Bridge Matching (2026 preprint)

# Beyond Classical OT

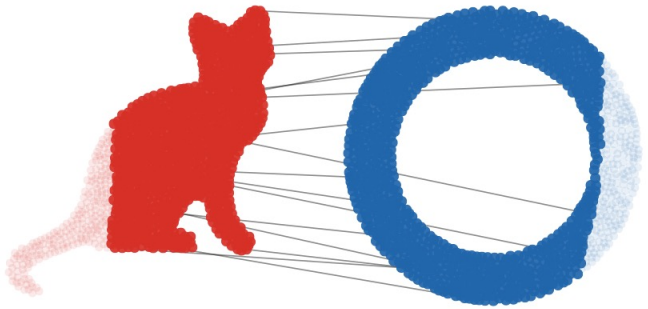


# Beyond Classical OT

## Which assumption should we relax?

Source

### Partial OT

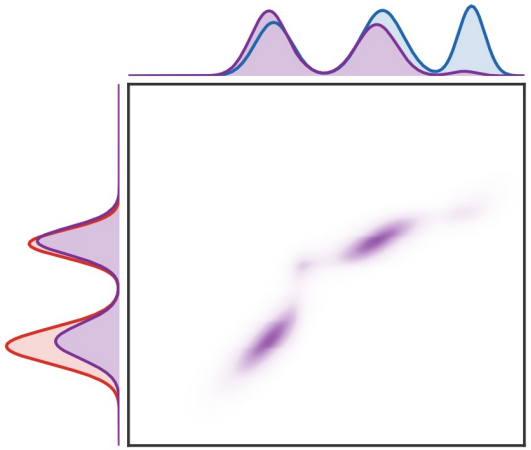


**Only part should match?**

$$\mathbf{1}^\top P \mathbf{1} = m$$

Occlusion and outliers

### Unbalanced OT

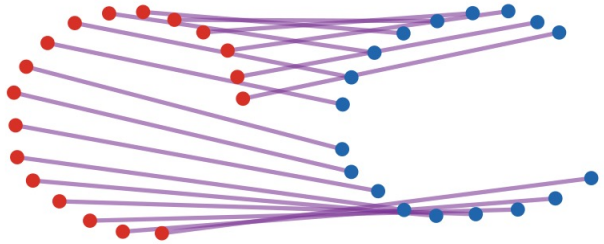


**Mass can change?**

$$(P\mathbf{1}, P^\top \mathbf{1}) \approx (a, b)$$

Changing populations

### Gromov–Wasserstein



**No common coordinates?**

$$D^X_{ik} \approx D^Y_{j\ell}$$

Different domains

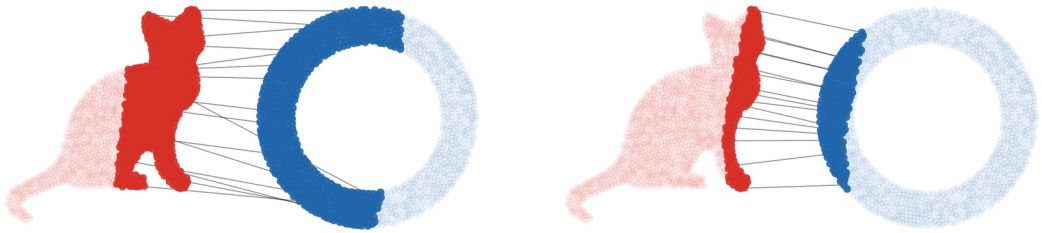
Figures: G. Peyré, OT4ML. The extensions can be combined in one transport model.

# Partial Optimal Transport

## Choose what to match, rather than forcing full coverage

### Select how much mass to match

**58% transported** **18% transported**



Saturated: matched. Pale: left unmatched.

$$\begin{aligned} \min_{P \geq 0} \quad & \langle C, P \rangle \\ \text{s.t.} \quad & P\mathbf{1} \leq a, \quad P^\top \mathbf{1} \leq b, \\ & \mathbf{1}^\top P \mathbf{1} = m. \end{aligned}$$

Also: positive–unlabeled learning (Chapel et al., 2020)

Shape examples: G. Peyré, OT4ML

### Image translation with partial OT

Handbags to shoes: retain colour and texture

**Input** **Balanced OT** $w = 1$ **Partial OT** $w = 8$

$$T_\sharp \mu \leq w\nu, \qquad w \geq 1$$

Translate every input, cover only part of the target.

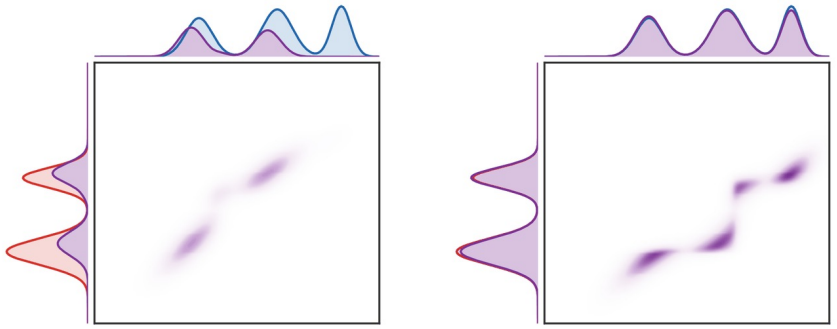
Gazdieva, Korotin et al., NeurIPS 2023

# Unbalanced Optimal Transport

## Pay for changing mass as well as moving it

### Soft marginal constraints

**Low mass penalty** **High mass penalty**



Red/blue: observed marginals
Violet: transported marginals

Low penalty: more local mass change.
High penalty: closer to given marginals.

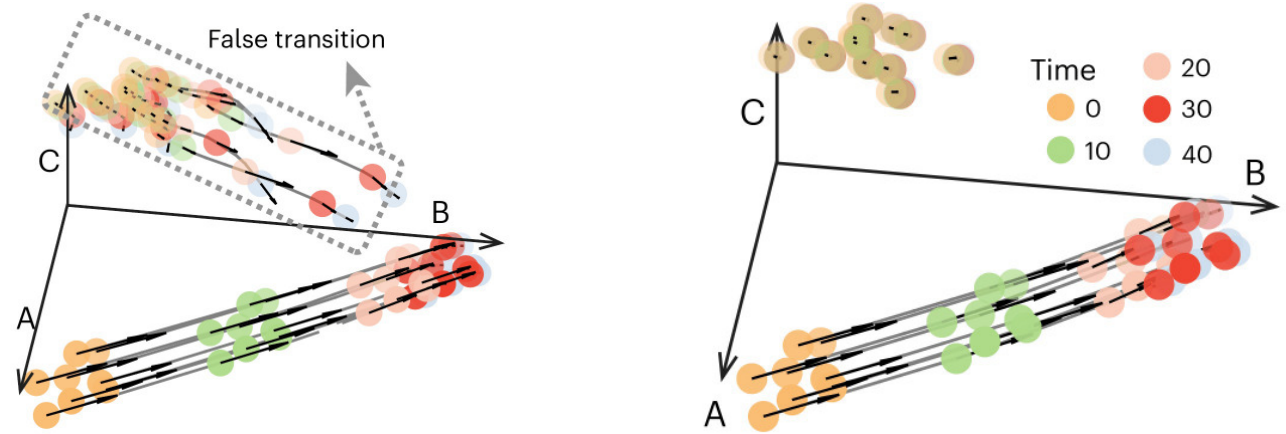
$$\min_{P \geq 0} \quad \langle C, P \rangle + \varepsilon \, \mathrm{KL}(P \| ab^\top) + \tau \big[ \mathrm{KL}(P\mathbf{1} \| a) + \mathrm{KL}(P^\top \mathbf{1} \| b) \big],$$

G. Peyré, OT4ML. Chizat et al. (2018).

### Cell fate inference with TIGON

Simulation: one group grows, another stays at rest

**Balanced OT (no growth)** **Unbalanced OT (TIGON)**




**Resting cells falsely move** **Resting cells stay put**

Local growth explains extra cells without inventing a state change.

$$\underbrace{\partial_t \rho + \nabla \cdot (\rho v)}_{\text{movement}} = \underbrace{g\rho}_{\text{growth or death}}$$

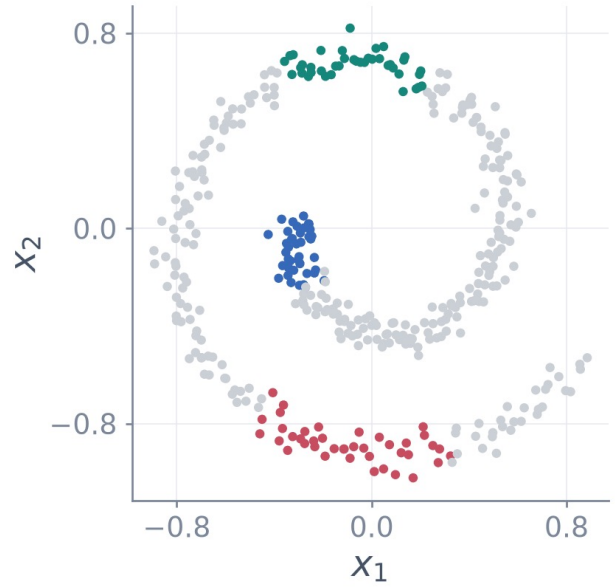
Sha et al., Nature Machine Intelligence 2024, Fig. 2b,g (CC BY 4.0)

# Gromov–Wasserstein Transport

## Match a 2D spiral to a 3D Swiss roll

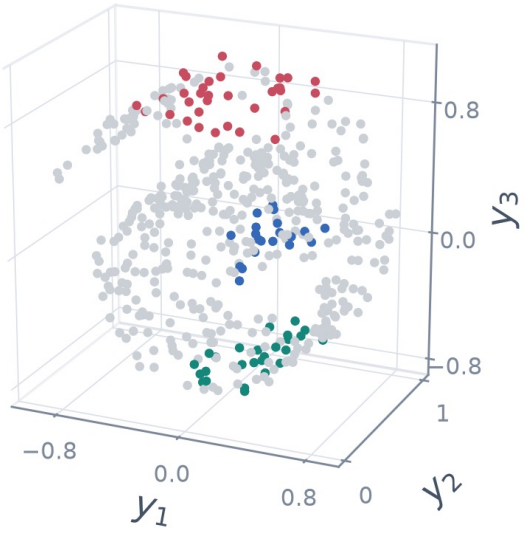
Source

### Source: 400 points in 2D




$P$

### Target: 500 points in 3D




Same colours mark strongest matches. Grey points also participate.

$$\min_{P \in \mathcal{U}(a,b)} \mathcal{L}_{\mathrm{GW}}(P), \qquad \mathcal{L}_{\mathrm{GW}}(P) = \sum_{i,k,j,\ell} \underbrace{\left|D^X_{ik} - D^Y_{j\ell}\right|^2}_{\text{relational distortion}} P_{ij} P_{k\ell}$$

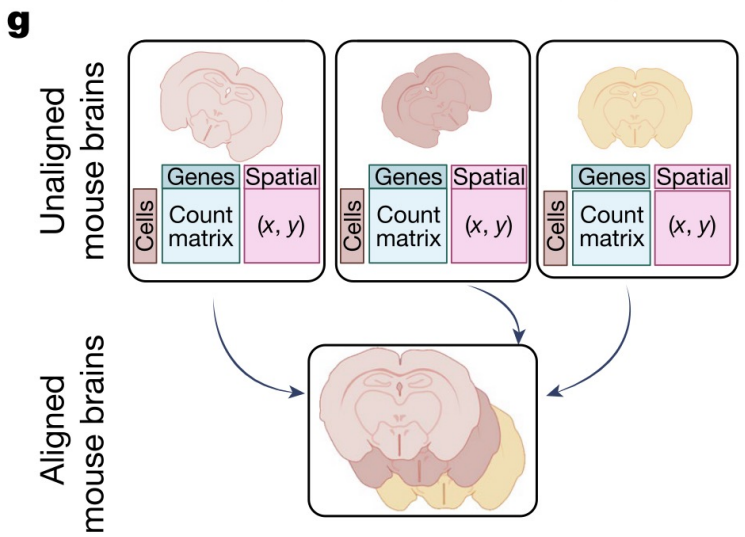
Generally non-convex. Symmetries can give several equally valid matches.

Coupling computed with POT. Mémoli (2011), Peyré et al. (2016).

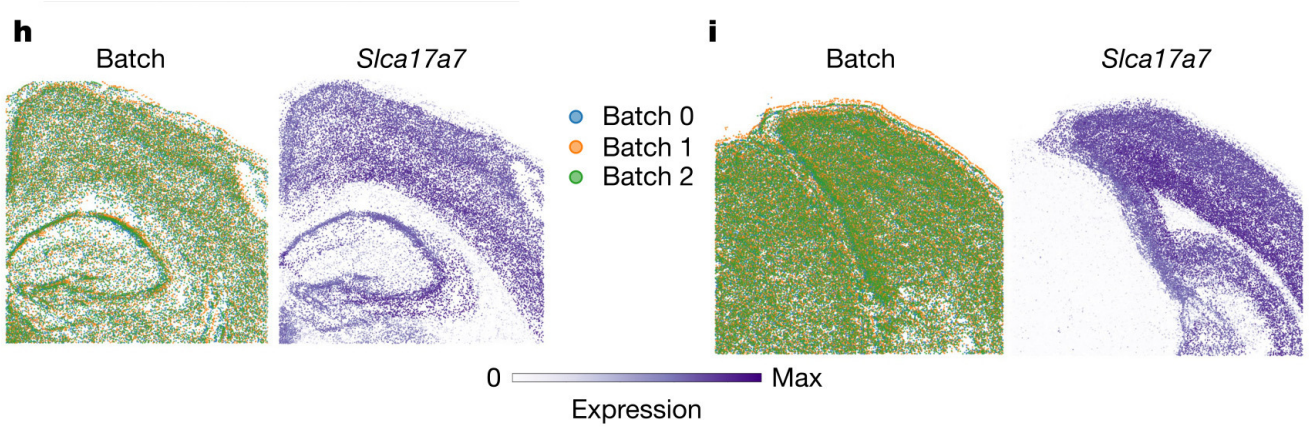
# Fused GW for Spatial Biology

## Match molecular identity and tissue geometry together

### A common frame for tissue sections




### Aligned mouse brain sections




Colours show sample batch and gene expression.

Gene expression identifies similar cells.
GW encourages a coherent spatial arrangement.

$$\min_{P \in \mathcal{U}(a,b)} (1-\lambda) \underbrace{\langle C_{\text{genes}}, P \rangle}_{\text{molecular similarity}} + \lambda \underbrace{\mathcal{L}_{\text{GW}}(P)}_{\text{spatial structure}}$$

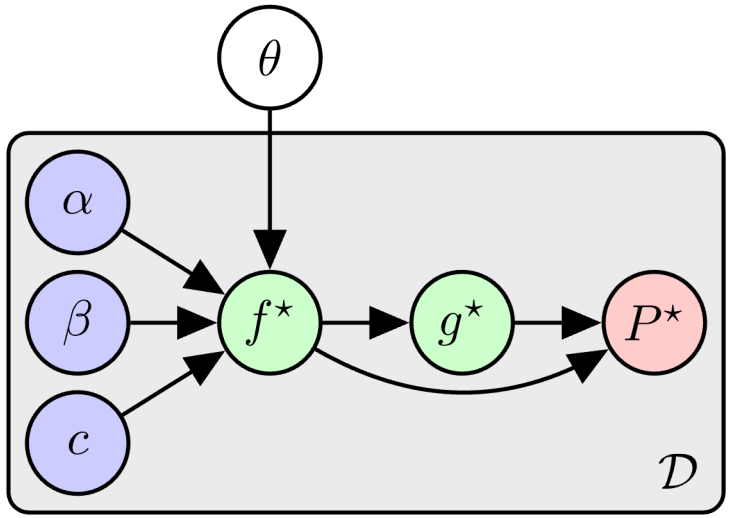
moscot, Nature 2025

Further: SCOT+, unbalanced cell/feature alignment (2026)

# Meta Optimal Transport

## Repeated transport problems share structure in their solutions

### Discrete: initialize Sinkhorn




Discrete (Entropic)

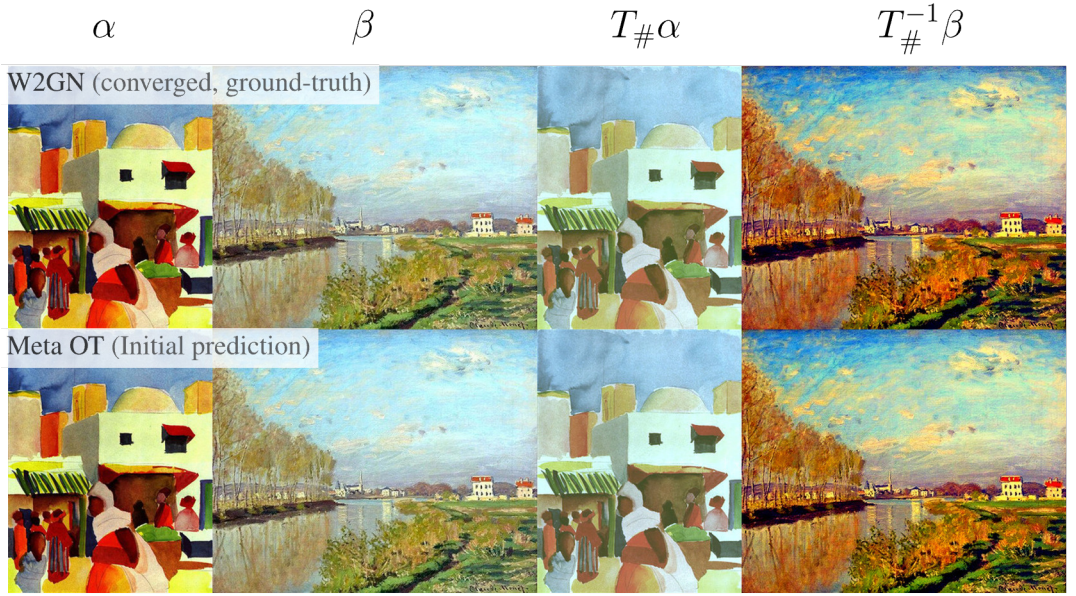
Predict approximate duals for each new pair.

$$(a, b) \longmapsto f_0 = \widehat{f}_\theta(a, b)$$

Refine with Sinkhorn to the required tolerance.

Amos, Luise, Cohen & Redko, ICML 2023

### Continuous: RGB colour transfer




Predict a convex potential for the RGB distributions.
The recoloured image keeps its spatial layout.

$$T_0(x) = \nabla_x \psi_{\widehat{\varphi}_\theta(\alpha,\beta)}(x)$$

OTT-JAX tutorial and code

# Meta OT: Learning and Refinement

## Train with the OT objective, without labelled optimal solutions

### Optimize the entropic dual

$$\mathcal{D}_\varepsilon(f,g) = a^\top f + b^\top g - \varepsilon \sum_{i,j} \exp\left(\frac{f_i + g_j - C_{ij}}{\varepsilon}\right)$$

$$f_\theta = \widehat{f}_\theta(a,b), \qquad K_{ij} = e^{-C_{ij}/\varepsilon},$$

$$g_\theta = \varepsilon \log b - \varepsilon \log\left(K^\top e^{f_\theta/\varepsilon}\right)$$

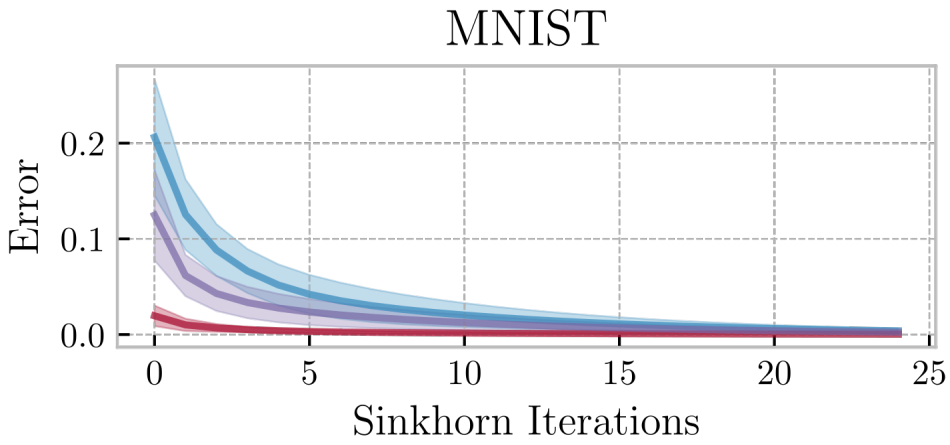
$$\boxed{\max_\theta \ \mathbb{E}_{(a,b)\sim\mathcal{D}}\left[\mathcal{D}_\varepsilon(f_\theta, g_\theta)\right]}$$

The dual value supplies the training signal.

Here the pixel grid, cost and regularization stay fixed.

Amos et al., ICML 2023, Sections 3.1 and 4.1

### Refine a prediction with Sinkhorn




**Zero init** **Gaussian** **Meta OT**

$$\mathrm{err}(P) = \|P\mathbf{1} - a\|_1 + \|P^\top \mathbf{1} - b\|_1$$

Here Meta OT starts with a lower marginal error.

OTT-JAX MetaInitializer